Human Re-Identification for Crowd Management

Final year project report submitted in partial fulfilment of requirement
for degree of Bachelors of Science in Electrical Engineering

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2026

DECLARATION

The project report titled “Human Re-Identification For Crowd Management” is submitted in partial fulfilment of the degree of Bachelors of Science in Electrical Engineering, to the Department of Electrical Engineering at Namal University, Mianwali.

It is declared that this is an original work done by the team members listed below, under the guidance of our supervisor “Dr. Mudassar Raza”. No part of this project and its report is plagiarised from anywhere, and any help taken from previous work is cited properly.

No part of the work reported here is submitted in fulfilment of requirement for any other degree/ qualification in any institute of learning.

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ACKNOWLEDGMENTS

First of all, we are thankful to Allah Almighty for giving us the strength, patience, and ability to complete this Final Year Project successfully.

We would like to express our sincere gratitude to our supervisor, Prof. Dr. Mudassar Raza, for his guidance, support, and valuable feedback throughout this project. His advice helped us improve our technical understanding and complete the project in a better direction. We are also thankful to our co-supervisor, Dr. Farrukh Qureshi, for his support and guidance during the project.

We are grateful to the Department of Electrical Engineering, Namal University, for providing us the opportunity to work on a practical and challenging project. This project helped us learn computer vision, machine learning, video processing, hardware acceleration, and real-world problem solving.

We would also like to thank our families for their constant prayers, support, and encouragement. Their motivation helped us stay focused during difficult stages of the project.

Finally, we thank all our friends and participants who helped us during video data collection and testing. Their cooperation made it possible to evaluate the system under different conditions.

This project was a valuable learning experience for us and helped us grow both technically and personally.

Abstract

This project presents the development and evaluation of a Human Re-Identification system using machine learning techniques for crowd management applications. The primary problem addressed is the inaccuracy of unique person counting caused by repeated counting when the same individual exits and re-enters the camera view.

The system integrates YOLOv8 for person detection, BoT-SORT for short-term tracking, and an OSNet-based re-identification model for appearance feature extraction. An identity gallery with cosine similarity and rank-based matching is used to determine whether a detected person has been seen before, and to assign or restore a global identity accordingly.

The OSNet-based model was fine-tuned using 11,335 cropped person images from a project-specific dataset of over 30 individuals. A query-gallery evaluation was also performed using 2,830 testing images and 35 query images. Optional face-assisted and supporting appearance cues are used when available. The Google Coral USB Accelerator was tested for compatible TensorFlow Lite feature extraction.

Results demonstrated that the system successfully detected, tracked, and re-identified persons in over a dozen recorded video scenarios including normal light, low light, occlusion and multi-angle same-camera conditions. The system reduced repeated counting in successful re-identification cases and provided CSV logs for identity decision verification. Limitations were noted under low light, heavy occlusion, similar clothing, and back-view conditions.

Chapter 1

Introduction

1.1 Background

Crowd monitoring is particularly relevant in areas frequented by crowds such as events, classes, meetings, seminars, exhibitions and public activities. Organisers may need to have an idea of the number of unique people who are present for various reasons, including managing seating, food, space, security, and other resources.

Generally, a basic system of counting a person's appearances in front of the camera would be used. This is a problem if the same person leaves the camera view and then comes back in. Even if the person has already been counted, the system can count him again as a new person. Therefore, the number of people might even be higher than what is actually reported.

This is called "re-identification failure." It implies that it is not being able to identify that a person is currently a detected person who has been detected before. This is not something that can only be resolved with tracking, as tracking typically only occurs when they are continuously visible. If the person is hidden from the camera's view or leaves the scene after some time, the track could be lost.

One attempt to solve this problem is human re-identification, which uses the appearance of a person being detected currently with the appearances of people that have previously been detected. If the person matches an existing identity, the identity is reinstated and the number of unique persons is not incremented. In this project, we will concentrate on machine learning for this task and generate a more accurate estimate of the number of unique individuals in a video of the crowd.

1.2 Project Context and Reason for Selection

The project was chosen because repeated counting is a real problem in crowd monitoring applications. In University events, seminars, exhibitions and indoor events, people tend to come in and out of the camera view a few times. These repeated sightings could be mistaken for multiple individuals by a normal detection/tracking system. This can result in a miscount of attendance and bad planning.

For instance, if visitors come and go from a building multiple times, the appearance-based counter will count a lot more than if everyone in the building was a different person. It includes seating arrangements, food service, crowd management and event planning. Hence, a system to recognise returning individuals would be more useful than a system that counts appearances.

This project was also chosen due to the ability to integrate computer vision, machine learning and evaluation of edge computing into one practical system. The system identifies persons, follows their movement as long as they can be seen, creates appearance features and matches these features with known identities. To enhance the appearance-based person identification in the project environment, an OSNet-based re-identification model was fine-tuned using about 15,000 cropped person images. Google Colab was used for model training and experimentation and the Google Coral TPU was tested for edge-based feature extraction and efficient local inference.

The project is relevant in that it is not just a cloud project. In many practical situations, internet connection might be restricted, unreliable or incompatible for transmitting video stream to the remote servers. The project benchmarks edge-based processing to suggest an alternative, more viable approach to controlled indoor crowd-monitoring.

1.3 Problem Statement

A primary issue tackled in this project is the inaccuracies in unique-person counting due to the same person showing up multiple times. A detection system could find people in a frame and a tracking system could track them for a few moments. When a person moves out of the camera's view and then comes back in, however, the system might identify the person as a different individual. This makes the count incorrect.

The problem is harder in real video as you can see people wearing similar clothes, walking in different directions, being lit in a different way, partial view, or back or side view. These conditions create a more difficult appearance matching. A reliable system would then need to match the face with the stored faces and determine if the face is new or if it has been previously seen.

Not only is the challenge detection of people, but it is also to keep a useful memory of the identities throughout the video. The system should eliminate double counting and should not double the number of instances where two persons are counted as one. This project will

tackle this issue by building a unique person counting system based on machine learning for human re-identification.

1.4 Aim and Objectives

Aim

This project aims to develop and deploy a human re-identification system based on machine learning techniques, which can detect individuals in crowd videos, track their appearance over time, identify re-appearing individuals even when they no longer appear in the camera field of view, and enable a more accurate count of unique individuals.

Objectives

1. To identify the presence of individuals in recorded crowd videos by an appropriate object detection technique.
2. To follow the detected people from one frame to the next when they are in the camera view.
3. To fine-tune an OSNet-based re-identification model using some 15,000 cropped person images and use the tuned model to obtain the appearance features for identity matching.
4. To compare extracted appearance features with stored identities to match returning individuals with their identity.
5. To minimize repeated count by only incrementing the unique count when a new person is found.
6. To test Google Coral TPU for edge-based feature extraction and effective local inference.
7. To assess the system in real usage scenarios, such as different lighting conditions, perspectives, hiding, and other types of clothing, from recorded videos.
8. To design the approach towards identity matching in a way that can be extended toward future person re-identification and counting in multi-camera scenario.

1.5 Overview of the Proposed Solution

The input of the system is a video of the crowd. The outputs include an annotated video, identity labels, a unique person count, and result logs. The system identifies persons in each frame and gives them temporary tracking IDs while they are present. Then, each person detected is cropped from the image and sent to the re-identification stage.

The re-identification stage involves extracting appearance features from each person detected. These are numeric expressions of appearance of the person. The extracted feature is then matched to the features stored of those who have been seen before in the video. If the current person is not distinct enough from a previous stored identity, the system will assign the stored identity again. If one cannot be found, it generates a new identity and adds to the unique count.

Appearance feature extraction is carried out with the OSNet-based model, due to the fact that person re-identification heavily relies on visual features. The model was further tuned using personal images of people cropped to project-specific images to better assist matching in the selected testing environment. The system also involves the use of ranking-based matching where the top few candidates of identity are matched before the final identity is decided.

The Google Coral TPU is tested on edge for feature extraction and efficient local inference. This phase examines how edge hardware can enable faster local computing and minimize reliance on cloud computing.

1.6 Criteria for Completion

The project will be judged complete when the following is accomplished:

1. The system is able to identify people in recorded crowd videos.
2. The system is capable of monitoring the detected people as they are continuously in view.
3. The system is able to extract appearance features from person crops detected from the image.
4. The system can remember characteristics of previously encountered persons.
5. The system can match a person it detects with a stored identity.

6. Previous identities can be given to a person who returns if there is a reliable match.
7. If the person does not correspond with any identity already stored in the system, the system can generate a new identity.
8. The system can generate an annotated output video with labels for identities.
9. The system can count the number of people without repeating counts.
10. The system is able to generate result logs to examine identity decisions and count performance.
11. The system can test Google Coral TPU for feature extraction at the edge or efficient local inference.
12. The system can be evaluated on video data with various practical scenarios like light changes, occlusion, different viewing angles, and different clothing.

1.7 Sustainable Development Goals and Sustainability Impact

This project is associated with the United Nations Sustainable Development Goals, particularly SDG 9 and SDG 11.

SDG 9 is about Industry, Innovation and Infrastructure. This project evaluates a practical crowd monitoring system using machine learning and edge-computing to support SDG 9. By using the Google Coral TPU, the project demonstrates a way to use AI that allows for more exploration near the source of the data than relying entirely on the cloud. This helps to enable more accessible and efficient technology within real environments like universities, halls and indoor event rooms.

SDG 11 is about Sustainable Cities and Communities. Precise crowd counting can assist organisations to better manage shared spaces. Resources including seating, food, staff and space can be planned more effectively if the number of unique persons is estimated more accurately. This can decrease the waste from over-estimated attendance.

The project also brings a sustainability advantage as it helps to limit the reliance on unnecessary cloud processing through local edge processing. The system is not designed to monitor individuals permanently; its goal is to identify and count unique persons during the project video session.

1.8 Scope of the Project

Recorded video footage of crowds is used to implement and evaluate this project in the context of controlled situations. The primary emphasis is on unique person counting by human re-identification. The system identifies persons, monitors them when they appear continuously, extracts appearance features, compares them with stored ones, and adjusts the number of unique persons.

The current deployment is done on recorded video footage, not on a large-scale deployment. The identity-matching approach is designed such that it can be extended to multi-camera person re-identification in the future, which is not part of the scope of this report.

Perfect recognition in all conditions is not claimed in the project. When people are heavily occluded, in poor lighting, wearing similar clothing, or when side or back views are available, performance can reduce. These limitations are inherent in real-world re-identification systems and are elaborated upon in the testing, results and discussion chapters.

1.9 Organisation of the Report

This report has nine chapters.

Chapter 1 presents the background, context and problem statement of the project, along with the aims and objectives, general solution, completion criteria, potential sustainability impact, scope and report overview.

Chapter 2 presents the literature review regarding person detection, tracking, human re-identification, appearance feature extraction, similarity matching and edge-based AI.

Chapter 3 explains the methodology followed for the development of the system. It presents the processing stages, the training of the models, the extraction of the features, the comparison of the identities, the testing of the edge devices and the constraints.

Chapter 4 describes the testing procedure. It describes the test videos, testing conditions, and the cases employed to test detection, tracking, re-identification, unique counting, and edge-device performance.

Chapter 5 reports the results of the project. It consists of the actual performance of the system, identity matching results, unique count output, result logs and the system behaviour under different conditions.

Chapter 6 presents and discusses the results with regard to the original problem. It describes areas of success, areas of challenge and why these challenges were encountered.

Chapter 7 concludes the report with a summary of the major project results and progress in unique person counting.

Chapter 8 presents future work, encompassing enhancements on real-time processing, larger datasets, more powerful re-identification models, and additional multi-camera deployment.

Chapter 9 offers reflections on learning, including technical skills, non-technical skills, teamwork, problem solving and personal growth acquired during the project.

Chapter 2

Literature Review

2.1 Human Re-Identification and Crowd Monitoring

Human re-identification, also referred to as person Re-ID, is the problem of determining if a person seen at a specific time and/or from a specific camera view is the same person as the one seen at an earlier time and/or camera view. This is a crucial task in crowd monitoring because the standard counting system in crowd monitoring can count appearances rather than individuals. A simple detection/tracking system will recount the number of people in the view of the camera if the person who left the view comes back. This results in the final count being higher than the actual number of people.

Person Re-ID is commonly seen as a matching/retrieval problem. A person image detected is used as a query and compared to a gallery of stored person identities. The ranks of the gallery identities are determined by the similarity between them and the system determines whether it is a new person or an existing person that the query belongs to. Ye et al. [1] summarized that current Re-ID systems consist of feature learning, metric learning, ranking and evaluation. This is closely tied with the current project as appearance features are stored, compared to the identities stored and the unique person counter is updated.

Re-ID is harder than just detection in practical crowd videos. Individuals can be in the foreground or background, with different lighting, angles, body positions, and occlusion. Meanwhile, various individuals can wear the same attire. These circumstances make it difficult to match identities. Hence, for a useful Re-ID system, the characteristics extracted from the image must be powerful enough to distinguish different people, and robust enough to identify the same person after a time delay.

2.2 Traditional Person Re-Identification Approaches

In the past, before the emergence of deep learning, Person Re-ID mainly relied on handcrafted features and metric learning. These techniques were based on colour, texture, local patterns, and descriptors of body regions for the representation of a person. The extracted features were then compared with some measure of distance or similarity.

Liao et al. [2] introduced a new set of features called Local Maximal Occurrence (LOMO) features and Cross-view Quadratic Discriminant Analysis (XQDA). LOMO was designed to be more invariant to viewpoint and illumination changes, and XQDA was designed to learn a discriminative metric for person image matching. It is significant work since it demonstrates two essential elements of Re-ID: appearance representation and similarity learning.

But conventional methods have obvious restrictions. Handcrafted features are not suitable for complex real-world videos. Their accuracy has been reduced when lighting changes, partial visibility, image quality is low, or when different individuals wear similar clothing. These weaknesses prompted the application of deep learning, in which the model is able to learn useful appearance features directly from training data.

2.3 Detection and Tracking in Re-ID Pipelines

The first step in a person Re-ID system is to detect the people in the video. Detection produces bounding boxes of persons per frame. The images are then cropped using these bounding boxes to track the persons and extract features from them. The quality of the detector bounding boxes matters since incomplete, blurred, or badly placed crops also affect the Re-ID stage.

The YOLO-based detectors are common in most practical video systems due to their fast processing speed and their ability to detect objects frame by frame. YOLO 2is a single-stage detection model that simultaneously predicts the bounding boxes and the class labels from the image. YOLO-based person detection is suitable for the current project because the system needs to first detect people before tracking and re-identification can be done.

Multi-object tracking is used after detection to provide short-term identity continuity. Tracking assigns temporal IDs to detected persons during the period of their appearance across consecutive frames. The ByteTrack [4] approach was to associate almost all the detection boxes rather than just high-confidence detection boxes.Motion information, appearance information, camera motion compensation, and an improved Kalman filter state representation were combined in BoT-SORT [5] to improve multi-pedestrian tracking.

Tracking can be helpful but not sufficient for unique person counting. A tracker could hold an ID when a person is always visible, but could lose an ID if the person goes outside the

camera's view, gets occluded, or comes back after a time. In the past these gaps have required re-identification to retrieve the pre-gap identity. Hence, for the current project, tracking is adopted for short-term continuity, and Re-ID is adopted for longer-term identity recovery.

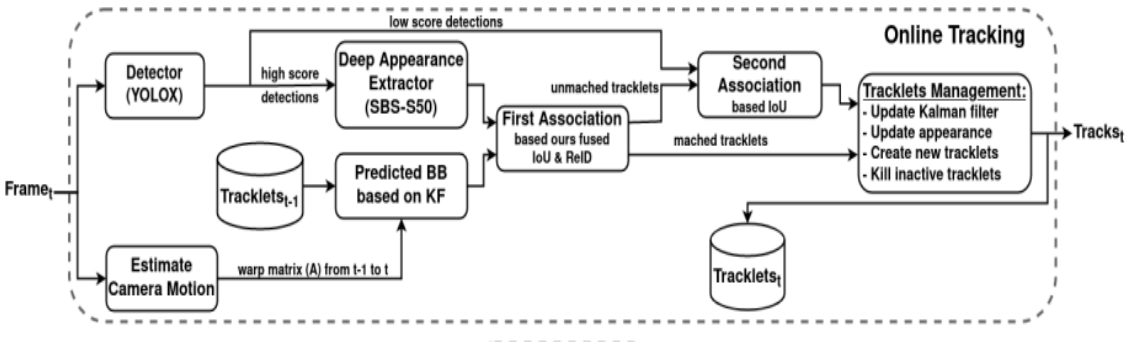


Figure 2.1: Multi-object tracking association process used to maintain short-term person identities across consecutive video frames using BoT-SORT [5].

2.4 Deep Learning-Based Person Re-Identification

The deep learning approach enhanced person Re-ID by enabling the model to learn appearance features from data rather than human-designed features. Every person image that is cropped is passed into a deep feature extractor which extracts a numerical feature vector from each image. This vector represents the image of the person. The system considers two crops to be similar if the feature vectors are close in the learned feature space.

Large Re-ID datasets were used to validate evaluation methods. Zheng et al. [6] proposed the Market-1501 benchmark to evaluate person Re-ID. It uses a query person image to be compared with a large gallery of candidate images and the system ranks the possible matches. This concept is also relevant for the current project as the system also looks for matches between the newly detected person and gallery identities.



Figure 2.2: Query-gallery matching concept in person re-identification, where a query person image is compared with stored gallery identities [6].

MARS [7] extended the single image Re-ID to video tracklets. The reason why video-based Re-ID is useful is that a person might not be clear in one frame but clear in another. But video-based models may be more complex and more costly in terms of computing. Image-crop-based feature extraction is more practical for this project.

Transformer-based Re-ID has also been investigated recently. He et al. [8] proposed TransReID, a transformer-based object Re-ID framework which adopted image patches and added further modules. Transformer-based methods are powerful but require higher computation and larger training resources. In this project, the lightweight CNN-based Re-ID model is more appropriate due to the need for balancing accuracy, speed, implementation time and hardware limitations.

2.5 OSNet for Appearance Feature Extraction

Omni-Scale Network (OSNet) was created specifically for person re-identification. Zhou et al. [9] proposed OSNet to learn omni-scale features for Re-ID. The key difference from feature extraction at a single spatial resolution is that OSNet employs multiple convolution streams to represent visual patterns at various scales. A common aggregation gate is used to combine these streams, enabling adaptivity in fusing multi-scale information in the network.

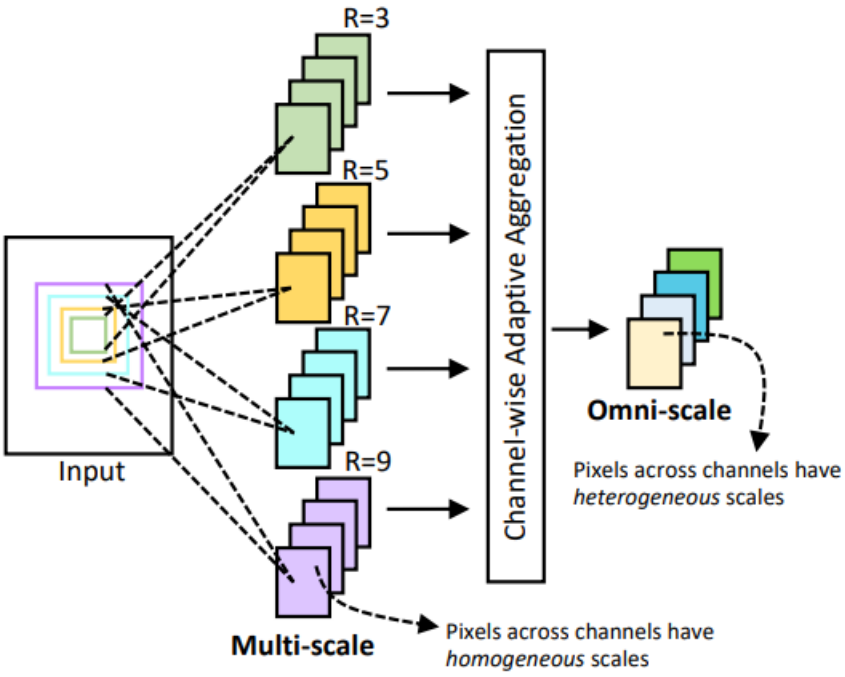


Figure 2.3: OSNet omni-scale feature learning block for person re-identification, comprising several convolution streams and adaptive feature aggregation [9].

This design can be valuable for person Re-ID as a person's identity may rely on both local and global visual clues. Local cues can be shoes, bags, texture in clothing, or colour patterns. Global cues can be the shape of the body or the overall appearance of full clothing. OSNet is lightweight yet captures both of them, in contrast to many large deep models.

Zhou et al. [10] later investigated generalisable omni-scale representations for person Re-ID. They found that learning features which are useful in other datasets and images under different camera setups were important. This is significant for practical systems as a model trained in one environment might not perform as well in another.

In the current project, OSNet is used as the main deep feature extractor for person appearance representation. Detected persons are cropped and fed to OSNet for extracting appearance features. All these features are then normalised and compared against stored gallery identities through similarity matching.

2.6 Fine-Tuning for Project-Specific Re-ID

Generalised Re-ID models, trained on public Re-ID datasets, might not be effective in every project environment. The recording angle, quality of the video, lighting, background, clothing style and other factors may differ from standard benchmark datasets. This gap can be minimized by fine-tuning the model, making it more suitable to the target environment.

In this project, the OSNet-based Re-ID model is fine-tuned with about 15,000 cropped person images collected from the project environment. The idea behind fine-tuning is not to develop a universal identity recognition system, but to optimize appearance feature extraction for the conditions set for testing. The fine-tuned model is applied in the person crop feature extraction process during video processing.

Fine-tuning is crucial as the project relies on the quality of the features extracted. The similarity matching stage can misidentify if the feature extractor yields unstable and weak features. Good features increase the likelihood of a returning person being correctly matched to the right stored identity.

2.7 Face-Assisted and Appearance-Based Matching

The most predominant cue in most videos of the crowd is body appearance since the whole body is typically more visible than the face. But it can be challenging to perform body-to-body matching with multiple people wearing the same clothes. In such instances, if the face is visible and clear, the information in the face can be used as additional support.

Face-assisted Re-ID is not a complete solution in this project, as faces are not always available, and should be used in conjunction with body-based matching. A person might be looking away from the camera, walking sideways, partly obscured or in low light conditions. Thus, face data should be used as a supplementary identifier for identity, not as the primary identifier.

This is the approach taken in the current project. Body features based on OSNet are the primary identity cues, with facial cues being helpful when available. This is the ideal design for crowd videos as it does not require the system to be fully dependent on the visibility of the faces.

2.8 Similarity Matching, Ranking, and Identity Gallery

Once the person is detected in the feature extraction system, the system has to determine if the person has been seen previously in the video. To this end, an identity gallery is used. Feature vectors of people who have been recorded in the gallery are stored. If a new detection occurs, the feature vector of the new detection is compared to the gallery features.

The most frequently used comparison method is cosine similarity, used to compare normalised feature vectors. The higher the cosine similarity, the more visually similar two features are. If the highest similarity score exceeds the threshold, then the person detected is given the previous identity. Once the score is below the threshold, a new identity is created.

Ranking is important since the top score is not always the best. The identity decision is not confirmed if the top two candidates scored very close to each other. A top-k ranking method is used to review the nearest identity candidates before selecting the final identity. The rank-based matching in the current project allows for more clearly understood identity decisions and the ability to analyze results from the logs.

The identity gallery directly enhances unique counting. Each time the system forms a new identity, the unique count increases. If a gallery identity is found during the matching process for a returning user, the previously assigned gallery identity is restored and the total number of unique identities is unchanged.

2.9 Edge AI and Coral TPU-Based Feature Extraction

Cloud-based video processing can cause delays, require a steady internet connection and pose privacy issues. By performing inference near the data source, Edge AI helps to minimize these problems. The benefits are helpful for crowd monitoring systems, as video data can be processed on the premises without continuous upload to a remote server.

The Google Coral TPU is an on-edge processor that enables efficient local inference with compatible TensorFlow Lite models. Seshadri et al. [11] tested the acceleration capability of Edge TPU for CNN and demonstrated that accelerators can enable edge inference for low-power applications, depending on the CNN architecture and compatibility. This is

important to the current project as the system tests an edge-based feature extraction method and efficient local inference using a Coral-compatible model.

It is crucial to set the Edge TPU role correctly. It is not guaranteed that the entire PyTorch OSNet and YOLO pipeline will be able to run directly on the Coral TPU. Rather, Coral TPU is used as an edge-inference evaluation with a model format compatible with TFLite for feature extraction. This ensures that the project scope is technically correct and not overstated in terms of the hardware deployment.

2.10 Research Gap and Relation to the Current Project

The literature reviewed indicates that person Re-ID has evolved from handcrafted descriptors to deep CNN-based feature extraction, OSNet-based representations, transformer-based methods and edge-inference research. Detection and tracking are essential components of a full video system but cannot resolve the issue of recognising a person who gets out of the camera's field of view and then returns.

The majority of person Re-ID studies concentrate on retrieval accuracy with benchmark datasets. This project, however, is mainly for the practical application of Re-ID into a unique counting problem within recorded crowd videos. The system should be able to detect human beings, track them if visible, extract appearance features using OSNet, compare the features with the identity gallery, update the unique count for identified persons, and assign the previous identity to persons if they return.

This project is therefore justified in five ways based on the literature. First, traditional Re-ID methods demonstrate the significance of appearance representation and similarity matching. Second, deep learning techniques enable the use of learned features rather than handcrafted descriptors. Third, OSNet offers a lightweight Re-ID-specific CNN architecture appropriate to extract features of cropped person images. Fourth, tracking methods including ByteTrack and BoT-SORT provide video identity continuity. Fifth, Edge AI literature supports the evaluation of local feature extraction with compatible accelerator hardware.

These studies give a technical basis for constructing a machine-learning-based unique person counting system for human re-identification. This project is oriented towards developing and testing a real recorded-video Re-ID system pipeline combining detection,

tracking, OSNet-based feature extraction, gallery matching, rank-based identity decisions and edge-inference evaluation.

Chapter 3

Methodology

In this chapter, the methodology followed in the design, training, implementation and evaluation of the proposed Human Re-Identification Using Machine Learning system is explained. The methodology is specific to the developed project, and provides a detailed explanation of the entire solution that is sufficient enough to allow the re-implementation of the system from the information presented.

The project integrates software video processing, machine learning model fine-tuning, re-identification logic and edge-based evaluation. The main software pipeline uses YOLOv8 for person detection, BoT-SORT for short-term tracking, an OSNet-based model for appearance feature extraction, cosine similarity for feature comparison, rank-based identity matching, and an identity gallery for unique person counting. The implementation also includes optional face-assisted and appearance-based supporting cues to improve identity decisions when a clear face or additional visual information is available.

The OSNet-based model was trained using 11,335 images from a dataset of 14,200 usable cropped person images gathered from over 30 persons. The model was trained for 60 epochs. The remaining images were divided into 2,830 testing/gallery images and 35 query images for query-gallery style identity matching evaluation.

The Google Coral TPU was tested for edge-based feature extraction and efficient local inference with a compatible TensorFlow Lite model. The project was validated with recorded videos obtained with the same camera setup but with varying viewing angles.

3.1 Design Process

The design process was created in a progressive way. Each stage is added in order to address a specific portion of the repeated-counting problem in crowd videos. A basic detection system can find people in a frame, but won’t know if the person was present in a previous frame. A tracking system can only maintain an identity as long as the person is always within sight. If the person disappears from the camera’s view or is obstructed or reappears after a period of time, the tracker can assign a different ID. This repeats the counting process.

To address this challenge, a novel design is introduced which integrates person detection, short-term tracking, appearance feature extraction, optionally used visual cues, identity matching, gallery updating and unique counting into one single pipeline.

3.1.1 System Requirements

The system has been designed to meet the following requirements:

1. Identify persons in video streams captured from crowds.
2. Follow moving objects between successive frames.
3. Detect person appearance features from person crops.
4. Rely primarily on body appearance cues of OSNet for identity.
5. Use face-assisted information, if a clear face is available.
6. Store features of previously seen people in an identity gallery.
7. Compare new detections with stored identities.
8. When the match is reliable, give the same global identity to someone who returns.
9. Only increment when a true new person is detected.
10. Create an annotated output video using identities and unique counts.
11. Create CSV logs with frame number, temporary track ID, global ID, similarity score, rank candidates, and identity decision.
12. Compare output of different query-gallery feature matching methods by returning Top-10 gallery images matching the query image.
13. Evaluate edge-based feature extraction using Google Coral TPU with a compatible TensorFlow Lite model.

3.1.2 Dataset Collection

The dataset was created from videos recorded in the project environment. The videos consisted of individuals walking, leaving the camera field of view, reappearing after some time delay, walking in another direction, and walking in different lighting conditions. These conditions were chosen as they represent the primary challenges in person re-identification and unique counting.

The recordings consisted of front, left, right, back, similar clothing, occlusion, normal lighting, and low-light views of the subject. A multi-angle same-camera setup was employed for the project. This means the same camera setup was used to capture persons at various angles and conditions. No multi-camera re-identification is claimed for the project as it is not evaluated with multiple separate cameras.

People who participated in the recordings have given consent. The system uses anonymous identity labels and unique counting for each video session. It does not make any correlations between identity and names, personal records or permanent biometric data.

3.1.3 Dataset Preparation and Organisation

Recorded videos were cropped to obtain person images for fine-tuning OSNet and re-identification assessment. Person crops were obtained from video frames using the detection stage. Bad quality crops, very blurry crops, incomplete crops, and misplaced regions were omitted if possible.

Over 30 persons were used to create 14,200 usable images of cropped persons. Different views of the same people, such as front, side, back, left view, and right view, were included in the dataset. The data was structured on the basis of identities. Images from the same person were grouped together with an identity tag. The final prepared dataset contained 11,335 training images, 2,830 testing/gallery images and 35 query images.

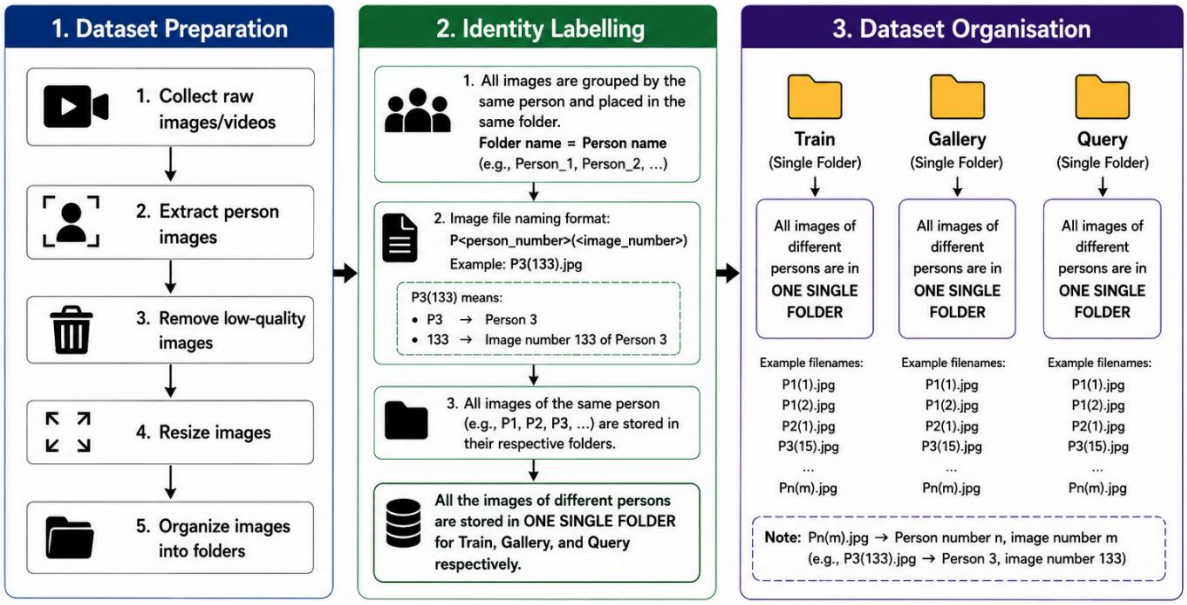


Figure 3.4: Workflow of Dataset Preparation, Identity Labelling, and OSNet Fine-Tuning Dataset Organisation

3.1.4 Preprocessing

Each person image was preprocessed to be suitable for the OSNet-based feature extractor before training and inference. Each crop was resized to a fixed input size and then converted to tensor and normalised. The same pre-processing was applied during both training and inference to ensure uniform input to the model.

The principal steps of the pre-processing were:

1. Trim the video frame and get the detected person region.
2. Change the crop size to the OSNet input size.
3. Convert to a tensor image.
4. Apply standard image normalisation to normalise the pixel values.
5. Do not use an unusable crop that is too small, blurred, dark or incomplete.

The final inference size used for the input was 256 × 128 pixels. This size was chosen because person re-identification models often operate on person crops in a tall rectangular shape which maintains the body appearance.

3.1.5 Data Augmentation

To enhance robustness and overcome overfitting, data augmentation was employed during training of OSNet. Realistic visual variation can make the model learn more stable appearance features, accounting for variation caused by lighting, viewing angle, image quality and body orientation. Augmentations were preserved in a realistic manner in order to retain all the images as person appearance. The goal was not to create fake identities, but to enhance the model's capability of identifying the same person across reasonable visual changes.

3.1.6 OSNet Fine-Tuning

OSNet was chosen because it is a person re-identification specific network. It learns to recognise appearance features in cropped images of people, extracting both local and global visual details. Local details could be shoes, bags, cloth texture, and minor colour patterns, while global details could include the shape of the body and clothing style.

To fine-tune the model, the OSNet-based model was trained on 11,335 images prepared from the project dataset in Google Colab. The usable dataset with complete data was 14,200 cropped images of persons representing over 30 identities. From these, 2,830

images were selected as the testing/gallery set and 35 images as query images for query-gallery style identity matching evaluation. The model was trained for 60 epochs. Once trained, the best learned models were stored and loaded in the final re-identification pipeline.

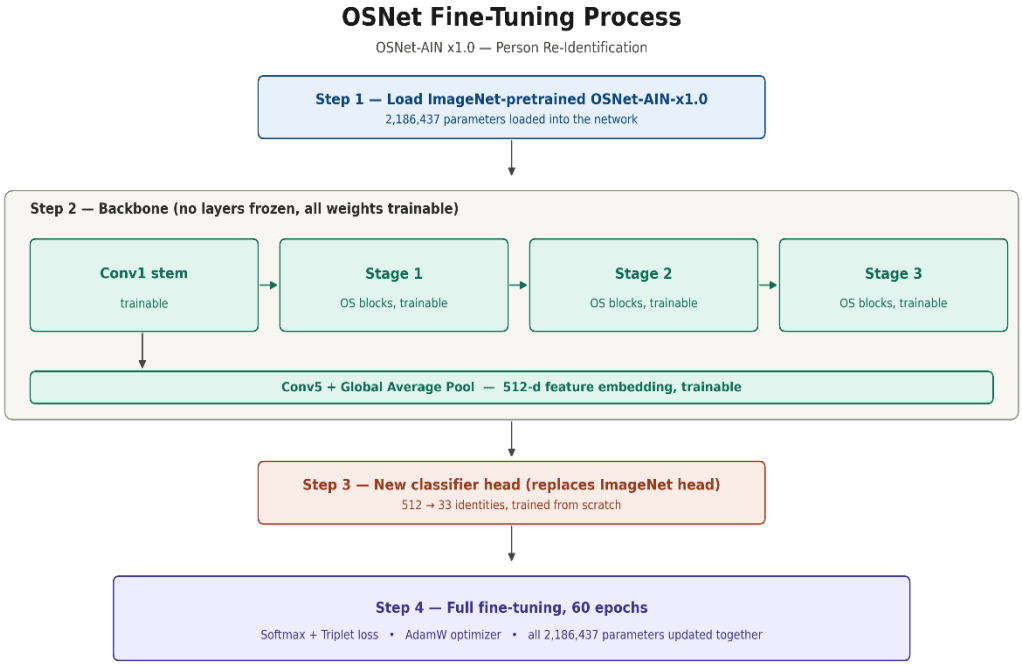


Figure 3.5: OSNet Fine-Tuning Process with 11,335 Training Images of Custom Person Dataset.

Table 3.1: OSNet Training Configuration

Parameter	**Value**
Base Model	OSNet (Omni-Scale Network)
Training Images	11,335
Testing/Gallery Images	2,830
Query Images	35
Training Platform	Google Colab
Number of Epochs	60
Input Size	256 × 128 pixels
Loss Function	Cross-Entropy with Triplet Loss

3.1.6.1 Query-gallery feature matching evaluation

A query-gallery feature matching test was also performed to check whether the trained OSNet-based feature extractor is able to find images of the same person from a larger gallery. This test had one query image compared to around 1000 gallery images. Cosine similarity between the feature extracted from the query image and the feature vectors of the gallery images was computed. The gallery images were then sorted by similarity score and the top ten most similar were shown.

This test was added to ensure the ability of the trained re-identification model to perform feature matching visually. The top ranked gallery images should be mostly from the same person as the query image if the model has successfully learned useful identity features.

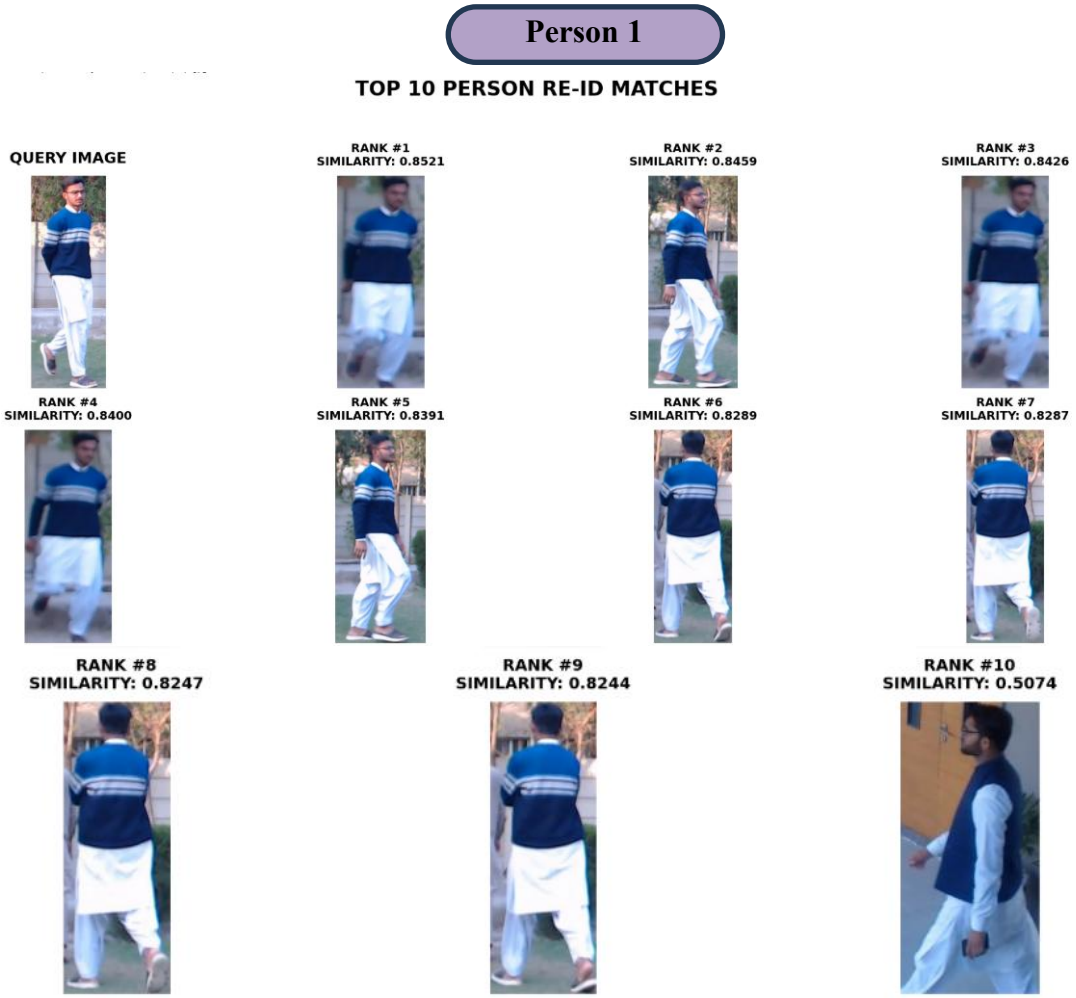


Figure 3.6: Query-Gallery Feature Matching Process with Top-10 Rank Image Retrieval from One Query Image.



Figure 3.7: Query-Gallery Feature Matching Process with Top-10 Rank Image Retrieval from One Query Image.

3.1.7 Detection, Tracking, and Re-Identification Pipeline

The main pipeline is based on a recorded video input. YOLOv8 is applied to each frame to detect people. The output of the detector is a set of bounding boxes around visible people. These detections are then fed into BoT-SORT, which gives temporary track IDs for humans appearing across frames.

The temporary tracking ID is not the permanent tracking ID. It is only used for short-term continuity. The final identity is assigned by the re-identification module through comparison of the person's appearance characteristics with the identity gallery.

The detected person is cropped from the frame and fed to the trained OSNet-based feature extractor. The model extracts an appearance feature vector from the crop. The feature vector is normalised and cosine similarity is used for comparing it with stored identity features. If they match an existing identity with enough confidence, the existing global ID is assigned. If a reliable match is not found, a unique ID is generated and the unique count is incremented.

Besides OSNet-based body features, face-assisted information is also optionally provided in the implementation when a clear face is available. This face cue is not used as the primary cue for identity since faces are not necessarily available in videos of crowds. Hence, the primary identity determination is made by the body appearance features in OSNet, while in appropriate cases, faces are employed as additional features to assist with identity determination.

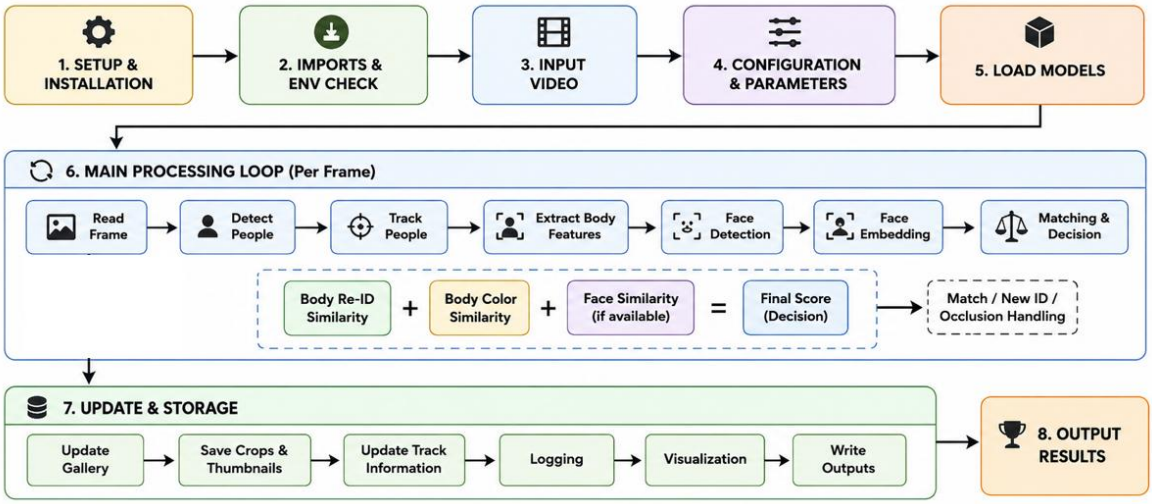


Figure 3.8: Overall Architecture of the Proposed Human Re-Identification System.

Table 3.2: Main Pipeline Configuration and Threshold Settings

Component	Method/Tool	Configuration
Person Detection	YOLOv8n	Confidence threshold: 0.4
Short-term Tracking	BoT-SORT	Default parameters
Feature Extraction	OSNet (fine-tuned)	Input: 256×128px, L2-normalised

Component	Method/Tool	Configuration
Similarity Measure	Cosine Similarity	Range: [0, 1]
Body Match Threshold	OSNet body-only	0.80
Fusion Match Threshold	Body + face fusion	0.82
Identity Gallery	Feature store	Updated on reliable match

3.1.8 Edge Hardware Evaluation

The Google Coral TPU was used to test edge-based feature extraction and efficient inference. This section of the project was not included as an assumption that a full YOLOv8 and PyTorch OSNet pipeline can be run on the TPU directly. The Coral TPU requires a compatible TensorFlow Lite model, which is quantised and compiled for the Coral TPU. The edge pipeline adopted a feature extraction model compatible with the Coral to perform local inference testing.



Figure 3.9: Coral TPU

3.1.9 Design Standards and Best Practices

There is no formal industrial design standard directly applied as it is an academic prototype. But good engineering practice was adhered to in design and implementation. The system was designed in a modular pipeline. The steps of detection, tracking, feature extraction, possible supplementary visual information, identity matching, counting, and hardware evaluation were divided up into individual steps. This facilitated the debugging, modification and evaluation of the system.

Reproducibility was taken into account by defining the dataset preparation process, the training split, the model used, the number of epochs, the input preprocessing, the matching

logic, the thresholds, and the output files. Privacy was taken into account by identifying with anonymous identity labels rather than names or personal data.

Table 3.3: Main Design Decisions and Justifications

Design Decision	Justification
YOLOv8 for detection	Fast, accurate single-stage detection suitable for video frames.
BoT-SORT for tracking	Robust multi-pedestrian tracking with improved Kalman filter.
OSNet for feature extraction	Lightweight Re-ID specific CNN with omni-scale feature learning.
Cosine similarity for matching	Normalised vectors allow direct similarity comparison.
Rank-based identity decision	More explainable and robust than single-score threshold.
Google Coral TPU evaluation	Tests edge deployment feasibility for local inference.

3.2 Mathematical Model

The core of the re-identification system relies on cosine similarity between normalised appearance feature vectors. For two feature vectors A and B extracted from person crops:

$$\text{Cosine Similarity} = \frac{A \,.B}{\|A\|\|B\|}$$

The similarity value ranges from 0 to 1. A value close to 1 indicates high visual similarity between the two persons, while a value close to 0 indicates low similarity. The threshold T is applied such that if the highest cosine similarity score exceeds T and the margin condition is satisfied, the returning person is assigned the previous global identity. Otherwise, a new identity is created.

For rank-based matching, the gallery is sorted by descending cosine similarity and the top-k candidates are evaluated. The margin condition requires that the best match score exceeds the second-best score by a defined margin M, reducing false matches between visually similar individuals.

3.3 Algorithm

The algorithm provides the logic for the system to detect a person, identify that person, and increment the unique count when appropriate.

Input: Recorded video V

Output: Annotated video, global identity labels, unique person count, and CSV result log

Step 1: Initialization

a. Load the trained OSNet-based model.
b. Initialize the model directory.
c. Load the BoT-SORT tracker.
d. Initialize an empty identity gallery G.
e. Set unique counter C = 0.

Step 2: Read the input video frame-by-frame.

Step 3: For each frame:

a. Localize persons using YOLOv8.
b. Delete low-confidence or weak detections.
c. Transmit successful detections to BoT-SORT.
d. Give temporary track IDs.
e. Cut out every detected person from the frame.
f. Inspect crop quality.
g. Crop, resize and normalise.
h. Extract appearance feature vector using OSNet.
i. If a clear face is present, extract face information (optional).
j. Identify visual or lower-body cues when present.
k. Normalise the extracted feature using L2-normalisation.
l. Compute cosine similarity between the feature and every identity in the gallery.

m. Rank the closest gallery candidates.

n. If there is a best match that meets the similarity threshold and margin condition: use the same global ID as before; do not alter the unique count; update the gallery using the reliable feature.

o. Else: create a new global ID; incorporate the feature into the gallery; add 1 to the unique count.

p. Draw a bounding box, global ID and unique count on the frame.

q. Store the identity decision, similarity score, rank information, and supporting cues in the CSV log.

Step 4: Save the annotated output video.

Step 5: Save the final result logs.

Optional Query-Gallery Retrieval Step:

1. Choose one query image.
2. Extract its feature vector using OSNet.
3. Extract/load the feature vectors of approximately 1000 gallery images.
4. Compare the query feature with all gallery features using cosine similarity.
5. Arrange gallery images in descending order based on similarity score.
6. Show the top 10 images in the gallery.
7. Determine the number of retrieved images that are of the same person as the query image.

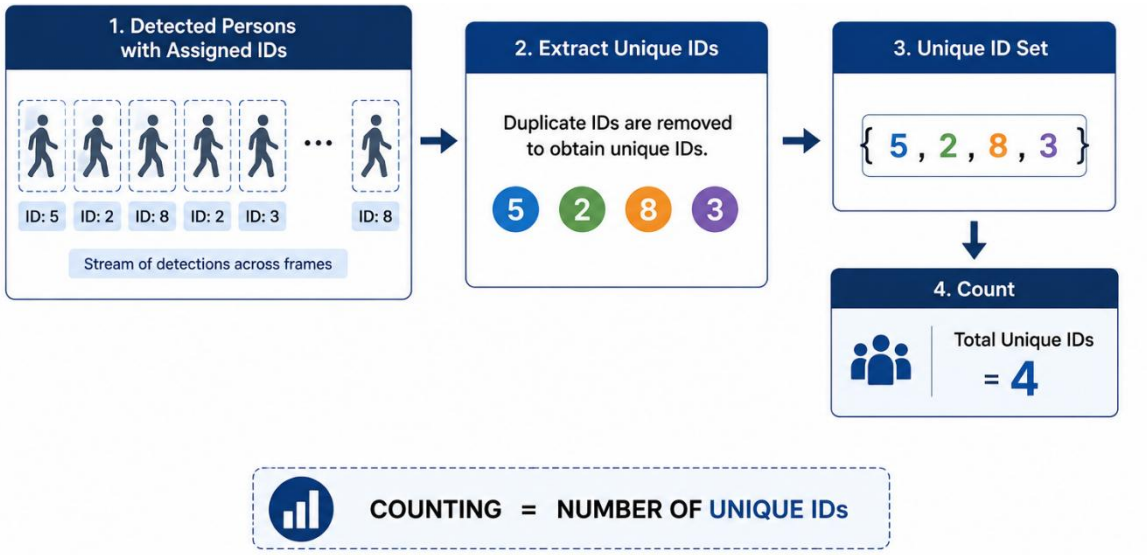


Figure 3.10: Flowchart of the Human Re-Identification and Unique Counting Algorithm.

3.4 Block Diagram

The system block diagram shows a full system of the proposed solution. The input is a video file of the crowd. The system detects, tracks, crops, preprocesses the video, extracts features using OSNet, collects face and appearance cues, matches with the identity gallery using the rank matching algorithm, and updates the unique count. The result is an annotated video and a CSV output file with identity decisions.

The main blocks included in the block diagram are:

1. Recorded video input
2. Frame extraction
3. YOLOv8 person detection
4. BoT-SORT tracking
5. Person cropping
6. Crop quality checking
7. OSNet-based feature extraction
8. Optional face-assisted feature branch
9. Appearance and lower-body feature branch
10. L2 normalisation

11. Identity gallery

12. Cosine similarity comparison

13. Rank-based identity decision

14. Unique count update

15. Annotated video output

16. CSV result log

17. Top-10 query-gallery retrieval branch (evaluation)

18. Coral TPU compatible feature extraction branch (edge evaluation)

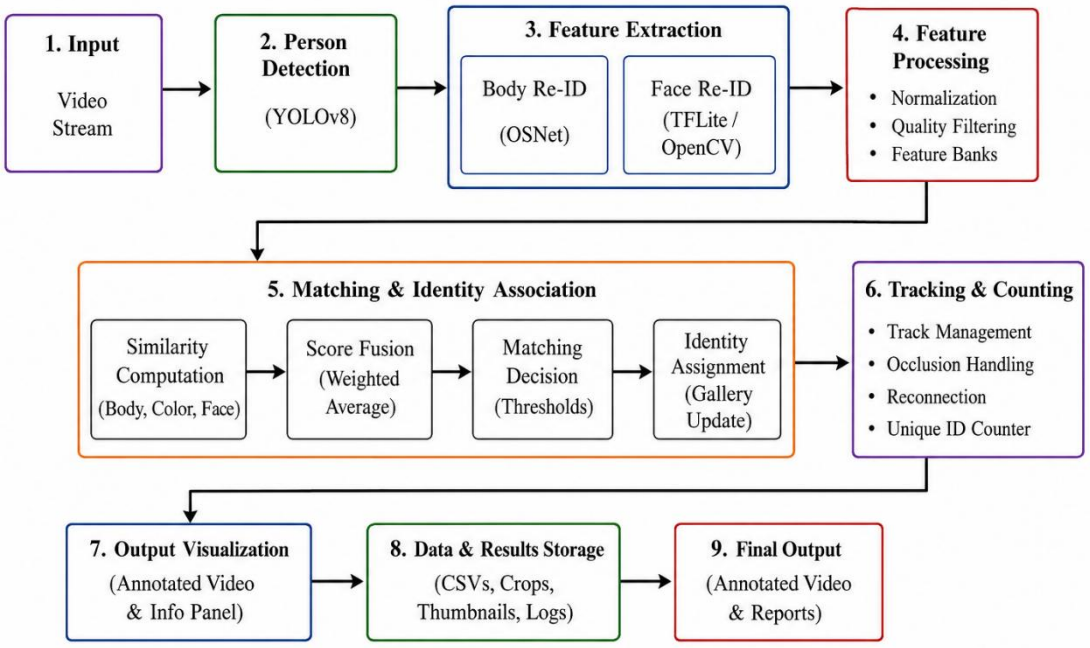


Figure 3.11: Block Diagram of the Proposed Human Re-Identification and Unique Counting System.

3.5 Implementation Constraints

There were some limitations on the implementation and functioning of the project. These constraints were acknowledged in the design, testing and evaluation.

3.5.1 Occlusion

When one person blocks another person in the frame, the detector might miss the person or give a partial bounding box, or the tracker might miss the person for a brief period. Occlusion may lead to loss of feature quality and may result in identity errors.

Measure Used: The system employed quality checking of the crops and did not update the gallery with unreliable crops. This avoided the loss of identity representations due to weak features.

3.5.2 Similar Clothing

Different people wearing similar clothing may produce similar features. If the similarity threshold is too low, this can lead to wrong identity matching.

Measure Used: Rank-based matching and margin checking was used in this system. The best identity candidate was only accepted if its similarity score exceeded the next identity candidate by a defined margin. When available, lower-body cues and face-assisted cues were also used to aid in identity decisions.

3.5.3 Face Visibility Limitation

Face information can be useful for re-identification but not in all crowd videos. The person can be viewed from the side, be partially visible or in dim light. Thus, relying on face information alone would result in an unstable system.

Measure Used: Face-assisted information was only used as an optional supporting cue. The primary identification task was still performed using OSNet body appearance features. When no clear face was available, the system ran without the face cue.

3.5.4 Lighting Variation

Low light and uneven lighting can cause images to be less clear and impact feature extraction. The same individual can look different in different frames if there is poor lighting.

Measure Used: Various lighting conditions were included in the testing of the system. Where possible, poor quality crops were filtered out and OSNet fine-tuning was used to enhance feature extraction in the project environment.

3.5.5 Different Viewing Angles

The same person can appear differently when viewed from front, back, side, left and right perspectives. This is challenging since Re-ID relies primarily on visual appearance.

Measure Used: Multi-angle same-camera samples were included in the dataset. The OSNet-based model was fine-tuned using these samples to learn features from various views of the same people.

3.5.6 Tracker ID Loss

While an individual is visible, BoT-SORT can preserve the ID number, but the temporary track ID could change if the individual leaves the frame and re-appears later.

Measure Used: Temporary tracker IDs were not used as permanent IDs. The final global IDs were generated through feature matching with the identity gallery by the OSNet feature matching engine.

3.5.7 Dataset Size and Diversity

The dataset comprised 14,200 usable cropped images from over 30 people. This was helpful for fine-tuning, but still restrictive in comparison to large-scale public Re-ID datasets.

Measure Used: Diversity was enhanced by gathering data in various views and practical situations. The restriction is still recognised since larger datasets can help improve generalisation.

3.5.8 Query-Gallery Test Scope

The Top-10 query-gallery retrieval test was performed with approximately 1000 gallery images and one query image. While this test proved helpful in visualising the feature matching capability, it was not a true test of accuracy with all identities and under all conditions.

Measure Used: This was considered supporting visual evidence in the feature matching process, not full benchmark accuracy. All assessment remained video-based, supported by CSV logs, CMC accuracy, mAP, and difficult-condition evaluation.

3.5.9 Threshold Selection

The similarity threshold determines if a person is matched to a known identity or gets a new one. A low threshold could combine two persons, and a high threshold could fail to match the same returning person.

Measure Used: During testing, threshold and margin values were modified. Identity decision quality was checked using rank-based matching and result logs.

3.5.10 Coral TPU Compatibility

The complete YOLOv8 and PyTorch OSNet pipeline cannot be directly executed on the Google Coral TPU. It demands a compatible TensorFlow Lite model which has been quantised and compiled for execution on the Edge TPU.

Measure Used: A model for feature extraction compatible with the Coral TPU was used. This was an edge-inference test, not full system deployment.

3.5.11 Multi-Camera Limitation

Currently, the project does not support multi-camera deployment. The videos were recorded with the same camera setup with various viewing angles. The identity-gallery design can be expanded to future multi-camera operation, but it is limited to the evaluation of same-camera recorded videos.

Table 3.4: Implementation Constraints and Adopted Solutions

Constraint	**Impact**	**Measure Used**
Occlusion	Partial crops, tracking loss	Crop quality check; gallery not updated with weak crops
Similar Clothing	Identity mismerging	Rank-based matching and margin checking
Face Visibility Limitation	Unreliable face cue	OSNet body features as primary cue; face optional
Lighting Variation	Weaker features	Multi-condition testing; fine-tuning on project data
Different Viewing Angles	Appearance variation	Multi-angle dataset with fine-tuning
Tracker ID Loss	Temporary ID change on reappearance	Global IDs from gallery matching, not tracker
Dataset Size	Limited generalisation	Diversity through multiple views and conditions
Threshold Selection	False matches or missed matches	Rank-based matching and margin condition

Constraint	Impact	Measure Used
Coral TPU Compatibility	Cannot run full pipeline on TPU	Compatible TFLite feature extraction only
Multi-Camera Limitation	Not cross-camera tested	Scoped as same-camera multi-angle system

The complete methodology of the proposed Human Re-Identification Using Machine Learning system was explained in this chapter. Detection is performed using YOLOv8, short-term tracking is done with BoT-SORT, feature extraction is done with an OSNet-based model, face-assisted and supporting appearance-based cues are optional, re-identification is performed using cosine similarity and rank-based matching, and unique counting is done with an identity gallery.

Chapter 4

Testing

The testing methodology and test cases to test the proposed Human Re-Identification Using Machine Learning system are explained in this chapter. This chapter aims to demonstrate the testing of the primary functions of the system in accordance with the project objectives and the requirements during the proposal phase.

It was tested to see if the system could detect people, track people from one frame to the next, extract features for each person, identify if any people re-appeared in subsequent frames, keep a count of each person, output CSV logs, run the Coral TPU feature extraction branch with an Edge TPU compatible TensorFlow Lite model, and perform query-gallery feature matching.

This chapter only describes the testing strategy and test cases. Results are presented in a separate chapter and are summarized numerically, graphically, via screenshots, and by comparison in the form of output and then interpreted.

4.1 Testing Methodology

In the testing methodology, the same sequence was used as in the implemented system pipeline. Testing was conducted by testing each of the major modules individually and finally testing the entire pipeline as an integrated system. This method was used to ensure that each component was functioning properly before checking the entire re-identification and counting procedure.

The testing process proceeded as follows:

1. Person detection testing based on YOLOv8.
2. Short-term tracking testing with BoT-SORT.
3. Person crop extraction and preprocessing testing.
4. OSNet feature extraction testing.
5. Identity gallery and similarity matching testing.
6. Rank-based identity decision testing.

7. Top-10 query-gallery re-identification testing.
8. Unique count testing.
9. Face-assisted support testing (optional).
10. Similar clothing testing.
11. Occlusion testing.
12. CSV log generation testing.
13. Difficult condition testing (low light, back-view, multi-angle).
14. Google Coral Edge TPU feature extraction testing.

Beyond the question of whether the program ran successfully, the main goal of testing was to determine if the system carried out the re-identification logic required. The testing was framed around the essential activity of the project, which is to know if a person who had left and returned was considered the same person or a new one.

4.2 Test Video Selection

Recordings of over ten videos were tested. These videos were purposely chosen to represent both normal and difficult re-identification scenarios. The videos were recorded with the same camera setup in different viewing situations. The testing was carried out using multi-angle videos recorded by the same camera, rather than a multi-camera setup.

The test videos used were under the following conditions:

1. Normal lighting.
2. Low lighting.
3. Occlusion.
4. Similar clothing.
5. Front-view person appearance.
6. Side-view person appearance.
7. Back-view person appearance.
8. Person leaving and returning.
9. Several people in a single frame.

10. Face-visible and face-hidden scenarios.

The test conditions were chosen because they are known to have a significant influence on the likelihood of human re-identification, such as lighting conditions, body angle to the camera, partial view, similarity of clothing, visibility of the face and re-appearance of the person.



Figure 4.12: Sample Frames from the Recorded Testing Video Under Different Conditions.

4.3 Testing Environment

The test was performed using recorded video files and the implemented pipeline in Python. This software pipeline leveraged the person detection model YOLOv8, the tracking model BoT-SORT, face-assisted support (optional), feature matching against an identity gallery, and CSV result logging.

Model training and feature extractor preparation was accomplished using Google Colab. Video-based testing and output of annotated videos were conducted using the local system. For the feature extraction part, the Google Coral USB Accelerator was tested separately with a TensorFlow Lite model compatible with the Edge TPU.

Table 4.5: Summary of the Testing Environment

Component	**Tool / Platform**
Person Detection	YOLOv8n
Short-term Tracking	BoT-SORT
Feature Extraction	OSNet (fine-tuned)
Model Training	Google Colab

Component	Tool / Platform
Video Testing	Local system (Python)
Edge Evaluation	Google Coral USB Accelerator
Edge Model Format	TensorFlow Lite (EdgeTPU compiled)
Output Format	Annotated video + CSV log

4.4 Functional Requirements Tested

The test cases were developed based on the main functioning of the project. A single important aspect of the solution was verified for each test case. The functional requirements tested in this chapter correspond directly to the system requirements defined in Section 3.1.1.

Table 4.6: Functional Requirements and Corresponding Test Cases

Requirement	Test Case
Detect persons in recorded videos	Person Detection Test
Track persons between consecutive frames	Tracking Test
Crop and preprocess person regions	Person Cropping and Preprocessing Test
Extract appearance features using OSNet	OSNet Feature Extraction Test
Re-identify a returning person	Re-Identification Test
Top-10 query-gallery feature matching	Top-10 Query-Gallery Test
Increment count only for new persons	Unique Count Test
Rank-based identity decision	Rank-Based Matching Test
Optional face-assisted matching	Face-Assisted Matching Test

Requirement	Test Case
Evaluate similar clothing cases	Similar Clothing Test
Evaluate occlusion cases	Occlusion Test
Evaluate low-light cases	Low-Light Test
Evaluate multi-angle same-camera cases	Multi-Angle Same-Camera Test
Handle multiple persons in one frame	Multiple People in One Frame Test
Generate CSV logs	CSV Log Generation Test
Evaluate Coral TPU feature extraction	Google Coral Edge TPU Test

4.5 Person Detection Test

What Was Tested

This test examined if the YOLOv8 detector was successfully able to detect visible humans in the recorded videos.

Why It Was Tested

The first step of the pipeline is the detection of persons. Tracking, feature extraction, re-identification and unique counting for a person cannot be done if the person is not detected. So, detection testing became a necessary prior to testing later stages.

How It Was Tested

The test videos were fed into the YOLOv8 detector to obtain the recorded results. The output frames were examined to determine if any bounding boxes were created around people that were visible in them. This test was conducted in videos with normal light, low light, multiple people and partial occlusion.

Expected Output

The detector should be able to produce bounding boxes for all visible persons and send these to the tracking phase.



Figure 4.13: Person Detection Test: Bounding Boxes Around Visible Persons.

4.6 Tracking Test

What Was Tested

This test verified if BoT-SORT was able to keep temporary track IDs for subjects as they were visible in consecutive frames.

Why It Was Tested

Tracking provides continuity over a short time. It is useful for the system to follow an individual when he or she is always in sight. The tracker ID is not considered a final ID as it can change after occlusion or reappearance.

How It Was Tested

Bounding boxes of detected persons were fed to BoT-SORT. The output video with annotations was examined to see if the temporary tracking ID stayed constant during transitions between successive frames.

Expected Output

A visible person should keep a temporary track ID during continuous appearance. The tracker could issue a new temporary ID if the person moves out of the frame and comes back in. If so, the final identity recovery should be performed by the re-identification module.



Figure 4.14: BoT-SORT Tracking Test Showing Temporary Track ID Assignment.

4.7 Person Cropping and Preprocessing Test

What Was Tested

This test confirmed if the person regions detected from the image could be correctly cropped and prepared for feature extraction.

Why It Was Tested

The OSNet feature vector is directly affected by the quality of the crop. The identity feature becomes weak when the crop is too small, blurred, incomplete or when the background is large.

How It Was Tested

The person regions in the video frame were cropped based on the detected bounding boxes. The cropped images were visually verified to ensure that the body part of the person was present. Poor crops were filtered prior to feature extraction where possible.

Expected Output

The system should create usable person crops for valid detections and not use very poor quality crops in updates to the identity gallery..

4.8 OSNet Feature Extraction Test

What Was Tested

This test was run to see if the trained OSNet-based feature extractor could extract feature vectors from valid person crops.

Why It Was Tested

To re-identify, one needs to compare feature vectors. If feature extraction does not succeed, the system cannot carry out identity matching.

How It Was Tested

The trained OSNet-based model was fed accepted person crops. The system was tested to ensure that feature vectors were created and sent to the identity matching stage. The testing also showed that the feature extraction stage functioned with the final pre-processing format used in the pipeline.

Expected Output

For every valid person crop, a numerical appearance feature vector should be generated which can be compared with the gallery appearance feature vectors.

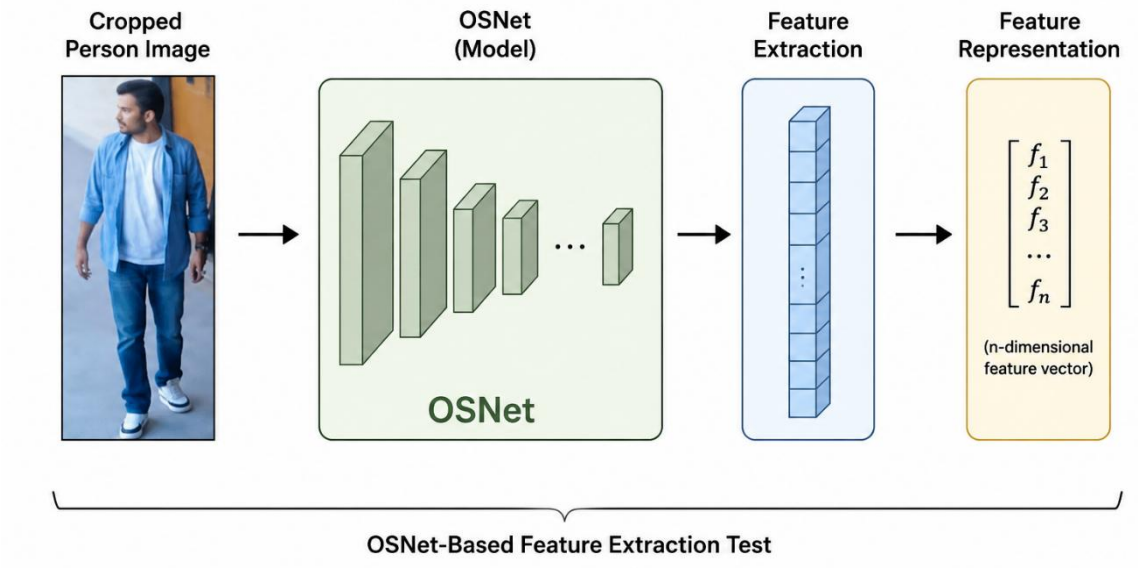


Figure 4.15: OSNet-Based Feature Extraction Test on Cropped Person Images.

4.9 Re-Identification Test

What Was Tested

This test was conducted to see if the system would be able to identify a person who went out of view of the camera and then came back.

Why It Was Tested

This is the main feature and operation of the project. If a person re-appears, a new ID may be given to them by a normal tracking system. The intent of the re-identification module is to match that person with the previous global identity.

How It Was Tested

Videos were chosen in which a person entered, exited the view and re-entered after a period of time. The system performed matching by cosine similarity and rank-based matching of the new feature vector to the identity gallery stored in the system. An annotated video and CSV log were reviewed to determine if the global ID was restored.

Expected Output

If the appearance match is reliable, the returning person should be given the previous global ID. If the match is not reliable, the returning person should be given a new ID.

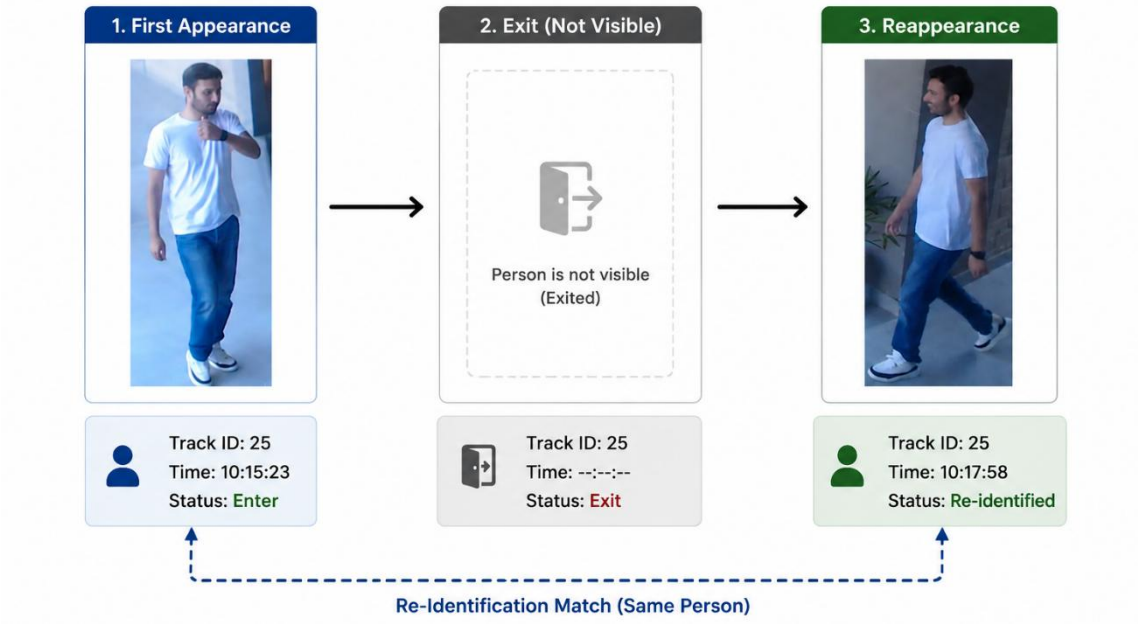


Figure 4.16: Re-Identification Test Showing First Appearance, Exit and Reappearance of the Same Person.

4.10 Top-10 Query-Gallery Re-Identification Test

What Was Tested

This test verified the performance of the OSNet-based feature extractor to retrieve images of the same person from a larger gallery when only one query image was provided.

Why It Was Tested

Feature matching is the main process used in the video pipeline. When a person is re-extracted, the most similar image of the same person should appear in the gallery if the model is strong. Hence, this test was employed to confirm the basic re-identifying capability of the trained model in a query-gallery format.

How It Was Tested

One query image was compared with around 1000 gallery images. A feature vector was extracted from the query image using the OSNet feature extractor, and feature vectors were also extracted from the gallery images. The query feature was compared to each gallery feature through cosine similarity. The gallery images were ranked and the Top-10 most similar were shown.

Expected Output

Most of the retrieved images should be from the same person as the query image. If the same identity is returned in the majority of the top-ranked positions, it is a strong result.

Observed Output

In the representative test for the query-gallery, the system returned practically all images of the same person in the Top-10 results. Between 9 and 10 of the top 10 retrieved images were the same as the query image. This demonstrated that the trained feature extractor could recognise the subject of the test in a larger set of images.



Figure 4.17: Top-10 Query-Gallery Re-Identification Test Showing Retrieved Same-Person Images for One Query Image.

4.11 Unique Count Test

What Was Tested

This test was performed to determine if the unique count was incremented only on the appearance of a new person.

Why It Was Tested

The goal of the project is to minimise repeated counting. Thus, unique counting is one of the most crucial tests.

How It Was Tested

Re-entry cases were included in videos that were processed through the full pipeline. The unique count displayed was verified on first appearance and reappearance of persons. The CSV log was also reviewed to ensure that the same global ID was not regenerated.

Expected Output

The unique count should go up if a new global identity is added. It should not increase if a person returns and is correctly matched to a past identity.

4.12 Rank-Based Matching Test

What Was Tested

This test determined if the system displayed the closest identity candidates correctly and used them accurately for matching.

Why It Was Tested

With rank-based matching, identity decisions can be analysed. It also aids in locating instances in which the best and second best candidates are nearly indistinguishable, suggesting doubt in the identity decision.

How It Was Tested

Test videos were processed and the CSV log was examined to find rank-based identity information. The best candidates, similarity values and final assigned identity were checked.

Expected Output

The system should return the highest ranked identity that it finds and apply similarity threshold and margin logic prior to accepting the match.



Figure 4.18: Rank-Based Identity Matching Test Showing Top Candidate Identities.

4.13 Face-Assisted Matching Test

What Was Tested

This test assessed the effectiveness of the optional face-assisted information to facilitate identity matching when the face was clearly presented.

Why It Was Tested

Some individuals may be wearing the same clothes making body-based matching harder. Face-assisted information may be helpful for matching in appropriate situations. But the system should not rely solely on visible faces as not all faces will be visible within crowd videos.

How It Was Tested

Video processing pipeline was used to process face-visible videos. The output and CSV logs were examined to determine if the face-assisted information was indeed available and used as a supporting cue. Face-hidden, side-view, and back-view videos were also tested to verify the system continued to use body-based OSNet features when face information was not available.

Expected Output

Face-assisted information can be used to assist with identity matching if a clear face is available. If no clear face is available, the system should continue to use OSNet body features.

4.14 Similar Clothing Test

What Was Tested

This test was conducted with various individuals wearing similar visual appearance clothing.

Why It Was Tested

Re-identification of persons wearing similar clothing is challenging because two non-identical persons can generate similar appearance scores. This can result in mismatch merging of identities if the system accepts a weak match.

How It Was Tested

Videos with people in the same type of clothing were processed. The rank candidates and similarity scores were checked in the CSV logs. The output video was checked to determine if the system assigned different identities or misassigned identities.

Expected Output

The system should employ ranking matching, margin checking and supportive cues to minimise wrong identity matching. This type of case is predicted to be harder than normal-light distinct-clothing cases..

4.15 Occlusion Test

What Was Tested

This test assessed system responses to the partial concealment of a person by another person or object.

Why It Was Tested

Depending on the duration of the occlusion, it can affect tracking and quality of crops. It can also cause difficulties in feature extraction due to the fact that the full body appearance might not be captured.

How It Was Tested

Videos with partial occlusion were processed. A test was made on the output video to determine if detections and temporary tracks were stable. The CSV log was examined to determine if occlusion led to uncertainty or incorrect identification.

Expected Output

The system should proceed to process under light occlusion. If the occlusion is more severe, there is a risk of tracking loss or unclear matching. The cases should be documented and openly addressed in the Results chapter.

4.16 Low-Light Test

What Was Tested

This test was conducted under low-light conditions to check the performance of the system.

Why It Was Tested

Low light decreases image clarity and can impact body feature extraction as well as face-assisted support.

How It Was Tested

The pipeline was used for processing low-light videos. The annotated output and CSV logs were used to verify detection quality, crop quality, identity matching and face availability.

Expected Output

Processing of low-light videos should be continued by the system, with a lower level of reliability compared to clear normal-light videos.

4.17 Multi-Angle Same-Camera Test

What Was Tested

This test assessed the ability to process multiple images of the same individual captured from different angles with the same camera configuration.

Why It Was Tested

The same person can look different when viewed from front, side or back. This poses a difficulty in the body-based re-identification aspect.

How It Was Tested

Videos were processed with respect to the front-view, side-view and back-view appearances. The system output was evaluated on its ability to deal with appearance changes using the identity gallery and OSNet-based matching.

Expected Output

The system should have higher performance if the body appearance is clear and consistent. Side-view and back-view cases are likely to be more complex, particularly when the face is not available.

4.18 Multiple People in One Frame Test

What Was Tested

This test determined if the system could handle more than one person per frame.

Why It Was Tested

When dealing with several individuals simultaneously, one needs to manage them correctly for crowd management. If the system is only for one person at a time, this is not appropriate for the project goal.

How It Was Tested

Multiple persons were visible in each video and were processed. The output video was validated to verify that each person got a bounding box, a temporary track ID, and an assignment of a global identity.

Expected Output

The system should recognise several people, track them separately, and assign global identities to the individuals through the re-identification logic.

4.19 CSV Log Generation Test

What Was Tested

This is a test to see if the system produces CSV logs during processing.

Why It Was Tested

The significance of CSV logs is that they enable the verification of identity decisions frame by frame. They also help check the result of the output video in addition to visually inspecting it.

How It Was Tested

The output folder was searched after running the pipeline for CSV log files. The log file was opened and checked to ensure that it had frame-level data, including the frame number, temporary track ID, global ID, similarity score, rank candidates, face availability, and bounding box coordinates.

Expected Output

The system should produce a readable CSV log of identity decisions and matching information for further analysis.

	A	B	C	D	E	F	G	H	I	J	K	
1	frame_idx	time_sec	processed_frame	method	track_token	assigned_id	is_new_id	assignment_source	face_available	rank1_id	rank1_score	rank2_id
2	1	0.033	1	body	1	ID_01	1	new	0		nan	
3	1	0.033	1	body	2	ID_02	1	new	0		nan	
4	7	0.233	1	body	1	ID_01	0	track	1	ID_01	0.9983174205	ID_02
5	7	0.233	1	body	2	ID_02	0	track	0	ID_02	0.9906813502	ID_01
6	13	0.433	1	body	1	ID_01	0	track	1	ID_01	0.995486021	ID_02
7	13	0.433	1	body	2	ID_02	0	track	0	ID_02	0.9860898852	ID_01
8	19	0.633	1	body	1	ID_01	0	track	1	ID_01	0.9978627563	ID_02
9	19	0.633	1	body	2	ID_02	0	track	0	ID_02	0.9972586036	ID_01
10	25	0.833	1	body	1	ID_01	0	track	1	ID_01	0.9974511862	ID_02
11	25	0.833	1	body	2	ID_02	0	track	0	ID_01	0.9955220222	ID_02
12	31	1.033	1	body	1	ID_01	0	track	1	ID_01	0.9966256022	ID_02
13	31	1.033	1	body	2	ID_02	0	track	0	ID_01	0.9939435124	ID_02
14	37	1.233	1	body	1	ID_01	0	track	1	ID_01	0.9960544109	ID_02
15	37	1.233	1	body	2	ID_02	0	track	0	ID_01	0.9934840202	ID_02
16	43	1.433	1	body	1	ID_01	0	track	1	ID_02	0.9935929179	ID_01
17	49	1.633	1	body	1	ID_01	0	track	1	ID_01	0.9950143099	ID_02
18	55	1.833	1	body	1	ID_01	0	track	1	ID_02	0.9961065054	ID_01
19	61	2.033	1	body	1	ID_01	0	track	1	ID_02	0.9890737534	ID_01
20	67	2.233	1	body	1	ID_01	0	track	1	ID_02	0.9879851341	ID_01
21	73	2.433	1	body	1	ID_01	0	track	1	ID_01	0.9946388006	ID_02
22	79	2.633	1	body	1	ID_01	0	track	1	ID_01	0.995931983	ID_02
23	85	2.833	1	body	1	ID_01	0	track	1	ID_01	0.9968479276	ID_02

Figure 4.19: CSV Log Output File Showing Frame-Level Identity and Matching Information.

4.20 Google Coral Edge TPU Feature Extraction Test

What Was Tested

This test was designed to see if the Google Coral USB Accelerator could be used and detected for feature extraction in the human re-identification system.

Why It Was Tested

The project has a hardware deployment component. This test aimed to validate the ability of the system to process an Edge TPU compatible TensorFlow Lite model for quick local feature extraction, rather than solely relying on CPU-based processing. This is crucial in crowd management applications that demand on-site processing and less reliance on cloud.

How It Was Tested

A Coral verification script was used to verify the Coral connection before running the video pipeline. Once the device was detected, the necessary Python libraries were verified such as PyCoral, TFLite Runtime, OpenCV, NumPy, Ultralytics and PyTorch. The compiled TensorFlow Lite model file was loaded into the Edge TPU compatible model: models/person_reid_edgetpu.tflite. The pipeline was executed with the Edge TPU argument and extractor option to guarantee that the system ran the Coral TPU feature extraction branch.

Expected Output

A system should detect the Coral TPU and execute without CPU fallback if an Edge TPU argument is provided.

Important Limitation

This test is not intended to imply that the entire YOLOv8 and PyTorch OSNet pipeline works on the Coral TPU. Only the compatible TensorFlow Lite feature extraction part was used with the Coral TPU. Detection and tracking remained part of the primary software stream.

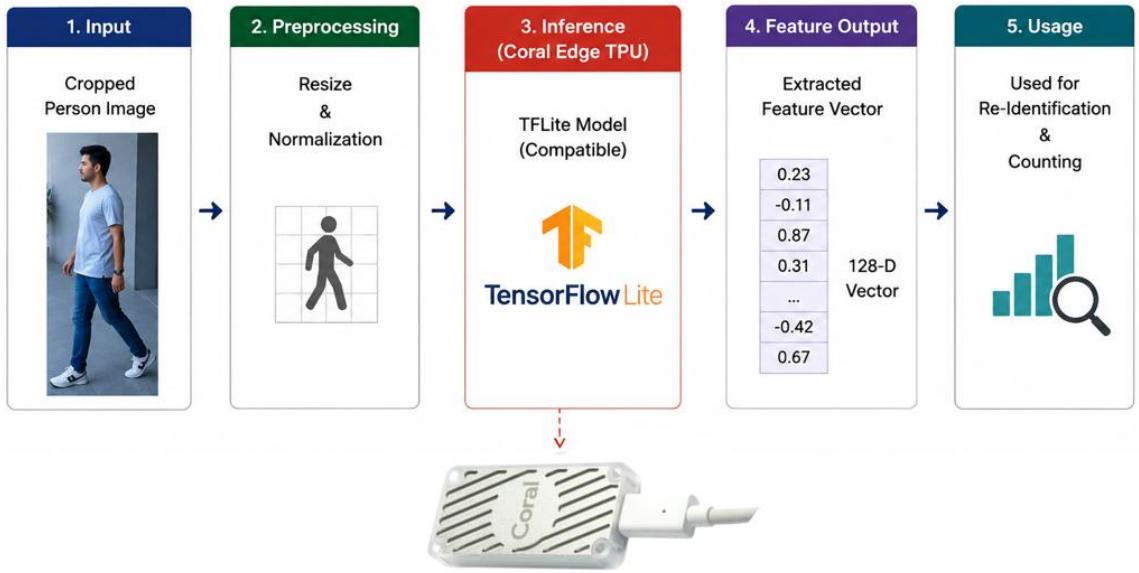


Figure 4.20: Google Coral Edge TPU Feature Extraction Test with a Compatible TFLite Model.

4.21 Overall Test Case Summary

Table 4.7: Summary of All Test Cases

Test Case	What Was Tested	Status
Person Detection	YOLOv8 bounding box detection	Tested
Tracking	BoT-SORT temporary track ID continuity	Tested
Person Cropping	Crop extraction and preprocessing	Tested
OSNet Feature Extraction	Feature vector generation	Tested
Re-Identification	Global ID recovery on reappearance	Tested
Top-10 Query-Gallery	Feature matching retrieval	Tested
Unique Count	Stable count for returning persons	Tested
Rank-Based Matching	Top candidates and margin logic	Tested
Face-Assisted Matching	Optional face cue usage	Tested
Similar Clothing	Identity matching under similar clothing	Tested

Test Case	What Was Tested	Status
Occlusion	Detection and tracking under occlusion	Tested
Low-Light	System performance in low light	Tested
Multi-Angle Same-Camera	Multi-angle same-camera re-identification	Tested
Multiple People in Frame	Multi-person detection and tracking	Tested
CSV Log Generation	Frame-level identity log output	Tested
Coral TPU Extraction	Edge TPU compatible feature extraction	Tested

4.22 Testing Summary

In this chapter, the testing methodology and test cases utilised to test and validate the proposed Human Re-Identification Using Machine Learning system are described. The testing covered the main functionalities of the proposal stage, such as person detection, tracking, OSNet-based feature extraction, re-identification, Top-10 query-gallery retrieval, unique count, rank-matching, optional face-assisted support, CSV logging, difficult condition testing and Google Coral Edge TPU feature extraction.

The system was evaluated with over a dozen recorded videos in normal light, low light, occlusion, similar clothing, front-view, side-view, back-view, person re-entry, and multiple-person conditions. The Top-10 query-gallery retrieval test was also conducted to visually evaluate feature matching by comparing one query image with approximately 1000 gallery images. The test results showed the Coral TPU could be utilised for the compatible feature extraction branch, whereas the whole YOLOv8 and PyTorch OSNet pipeline was not claimed to run directly on the TPU.

The detailed outputs, figures, training curves, CMC accuracy, mAP graph, Top-10 query-gallery retrieval result, unique count observations, success cases and failure cases are explained in the Results chapter.

Chapter 5

Results

In this chapter, the results from test cases described in Chapter 4 are presented. This chapter evaluates the implemented Human Re-Identification Using Machine Learning system by means of the important test cases applied during the evaluation. The results are supported by figures, plots, comparison tables, annotated video observations, and CSV log outputs.

The results concentrate on the most important results and trends that were seen in testing and do not display each single frame from each test video. Only the most significant output patterns are described in this chapter as over a dozen recorded videos were used. Complete frame-level logs and more screenshots can be added in the appendix.

5.1 Overview of Result Assessment

Recorded videos under various test scenarios including occlusion, person leaving/returning, multiple persons appearing in one video frame, person in different clothing, normal light and low light were successfully processed by the implemented system. Annotated output videos were created showing detected persons, global identity assigned and unique count. The system also produced CSV logs that can be used for verifying identity decisions and similarity scores.

From the testing the main result noted was that the system performed best under the conditions of clear body visibility, normal lighting and clothing difference between people. In these conditions, the system was able to minimise re-counting in representative re-entry situations by assigning the same global ID to a re-entered person. Performance was more challenging under low light, occlusion, similar clothing, back-view appearance, and hidden-face conditions.

Table 5.8: Summary of Major Result Areas

Result Area	**Performance Observed**
Person Detection	Good under normal light; reduced in low light and heavy occlusion

Result Area	Performance Observed
Short-term Tracking	Stable during continuous visibility; temporary ID may change on reappearance
OSNet Feature Extraction	Reliable features under clear visibility; weaker with poor crops
Re-Identification	Successful in normal conditions; challenging under similar clothing, back-view
Unique Counting	Correctly stable when returning persons are re-identified
CSV Logging	Consistent frame-level logs generated for all test videos
Coral TPU Extraction	Compatible feature extraction branch successfully executed

5.2 Dataset and Model Evaluation Results

The OSNet-based feature extraction model was fine-tuned with the prepared project dataset. There are 14,200 usable images of cropped persons from over 30 people. Of these, 11,335 images were selected for training, 2,830 for testing/gallery and 35 for query.

The query-gallery test was performed to see if the trained model can correctly identify a query image from a gallery of stored images. This evaluation is important as the final re-identification pipeline relies on comparing a new person crop with those already stored as identity features.

Table 5.9: Data Split for Training and Testing

Split	Number of Images	Purpose
Training	11,335	OSNet fine-tuning
Testing / Gallery	2,830	Query-gallery evaluation
Query	35	Representative query-gallery matching test

Split	Number of Images	Purpose
Total	14,200	Usable cropped person images

5.2.1 Training Loss Result

The training loss curve demonstrates the learning process of the OSNet-based model over 60 epochs. The loss was high at the beginning and decreased sharply during the early epochs. The curve after the initial drop became stable, indicating that the model was beginning to converge.

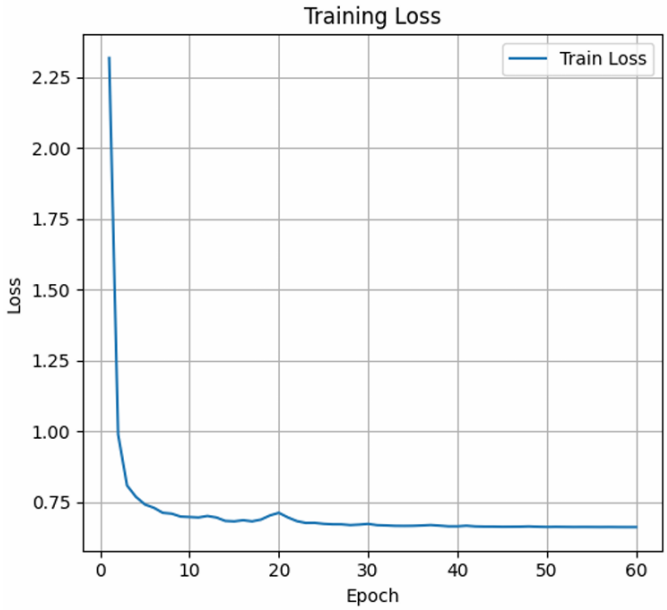


Figure 5.21: Training Loss Curve for the OSNet-Based Re-Identification Model after 60 Epochs.

The training loss result shows that the model has been able to learn some beneficial identity characteristics in the training images. The curve dips sharply in the early epochs and stabilizes in the later ones, indicating rapid learning during the initial epochs and a more consistent learning process in later epochs.

5.2.2 Mean Average Precision Result

Mean Average Precision was used to evaluate the retrieval quality of the trained feature extractor. The mAP provides a more general measure of retrieval performance over the query-gallery set, whilst CMC tests if the correct identity is present at a particular rank.

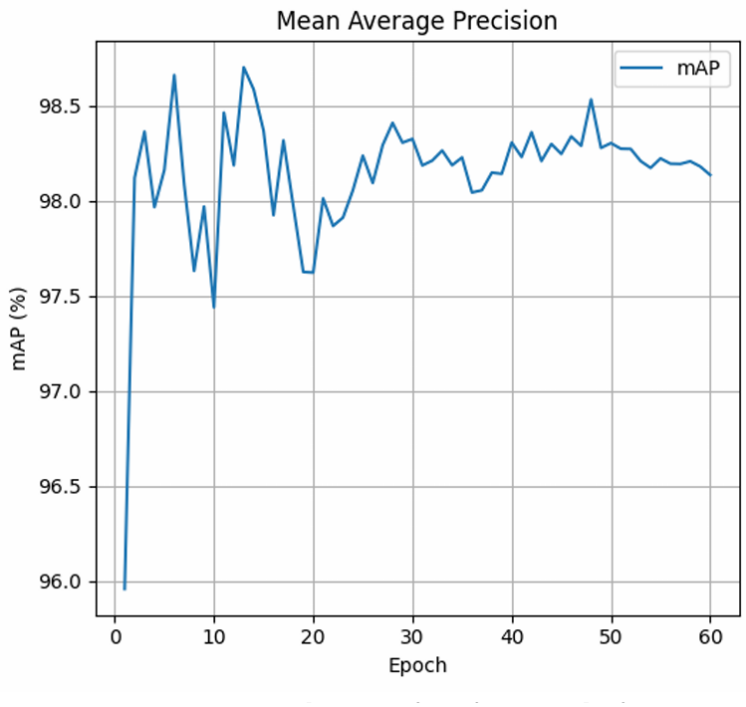


Figure 5.22: Mean Average Precision of the Fine-Tuned OSNet-Based Model During Query-Gallery Evaluation.

The mAP performance was satisfactory during the evaluation. Results indicate that the trained OSNet-based model achieved good identity retrieval accuracy on the prepared dataset. Small variations were seen from epoch to epoch due to variations in image angle, lighting, similarity of individuals, and query image quality.

5.2.3 Top-10 Query-Gallery Re-Identification Result

The trained OSNet-based model was tested in terms of the feature matching ability by showing a representative Top-10 query-gallery retrieval result. This test was conducted with about 1000 images in the gallery and one query image. The gallery images were sorted by cosine similarity and the Top-10 most similar images from the gallery were shown..

It was found that nearly all of the Top-10 retrieved images were from the same person as the query image. There were about 9–10 images in the Top-10 that had a matching identity. This provides visual evidence that the trained feature extractor was able to learn discriminative identity features for the tested person. This result is significant since it directly contributes to the re-identification logic implemented in the video pipeline.

Table 5.10: Top-10 Query-Gallery Retrieval Result Summary

Metric	Value
Total gallery images	~1000
Query images used	35 (representative test shown)
Top-10 same-person retrieved images	9 to 10 out of 10 (representative case)
Similarity measure	Cosine similarity

5.3 Person Detection and Tracking Results

The person detection test revealed that persons could be detected in the videos with YOLOv8. Bounding boxes were created for persons in normal-light, multiple-person and most clear-view scenes **(as shown in Figure 4.2).**

The person's visibility and the level of light were important factors in detection, with the clearer the person was visible and the more normal the light, the more likely detection was successful. In some instances, low-light frames, partial occlusion, and frame-edge positions compromised the detection quality.

In order to assign temporary track IDs, BoT-SORT was used after detection **(as shown in Figure 4.3).** The tracking result demonstrated short-term continuity of the identities of individuals who did not move outside the camera view across consecutive frames

The tracking result showed that the tracker ID is a useful identifier for continuous visibility, but is not a final identifier. If the person left the camera view, or was obstructed, the temporary track ID may change once they reappear. This validated the requirement for the OSNet-based re-identification module and identity gallery.

5.4 Re-Identification and Unique Counting Results

The primary goal of the project was to decrease repeated counting by being able to re-identify a person who left and came back. Videos where a person could be seen, then moved away from the camera and then came back after a period of time were used to verify the re-identification result **(as shown in Figure 4.5).**

The system matched the new feature vector to identity features stored in the gallery. If the similarity score and the margin condition were reliable, the previous global ID was used.

The results indicated that in successful cases, the system was able to give the same global identity to a person who returned to the system. This minimised repetitive counting since the unique count would not have been incremented when the individual was properly identified as being known.

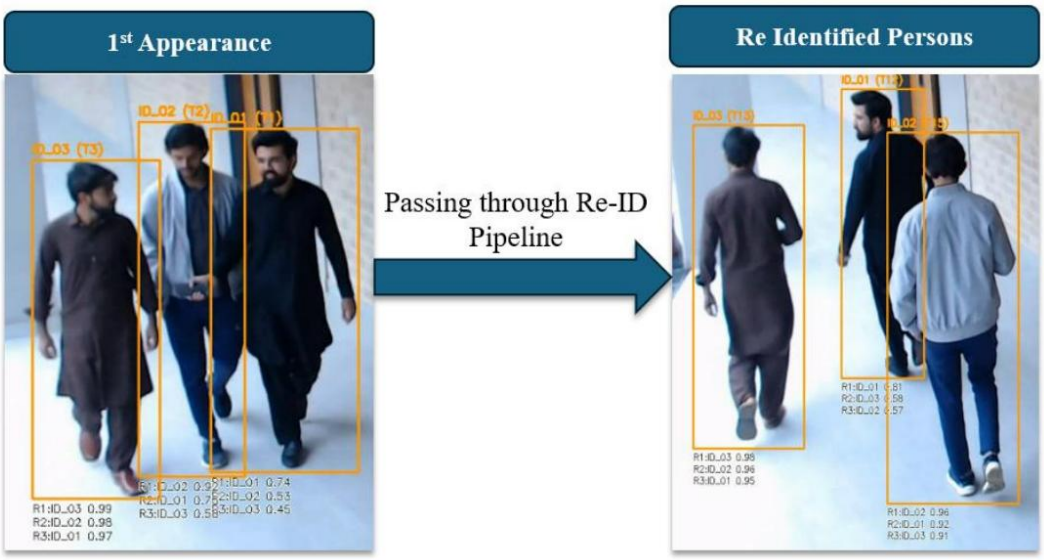


Figure 5.23: Unique Count Result Displaying Stable Count After Correct Re-Identification.

This confirms the difference between normal detection-based counting and re-identification-based counting. The former just detects the presence of an individual, and the latter tries to detect the identity of an individual by storing and comparing identity features.

Table 5.11: Unique Count Results Under Major Test Conditions

Test Condition	**Unique Count Behaviour**
Normal light, distinct clothing, front-view	Count stable on correct re-identification
Low light	Some repeated counting due to weak features
Occlusion	Occasional tracking loss; possible new ID assignment
Similar clothing	More difficult matching; some identity merging possible

Test Condition	Unique Count Behaviour
Back-view / side-view	More uncertain matching; increased new ID assignments
Multiple persons in frame	Each person tracked and assigned separate global ID

5.5 Rank-Based Identity Display Results

To demonstrate the closest identity candidates in the matching process, rank-based matching was employed. The system did not take the best match without considering the candidates' similarity margin and the top-ranked ones.

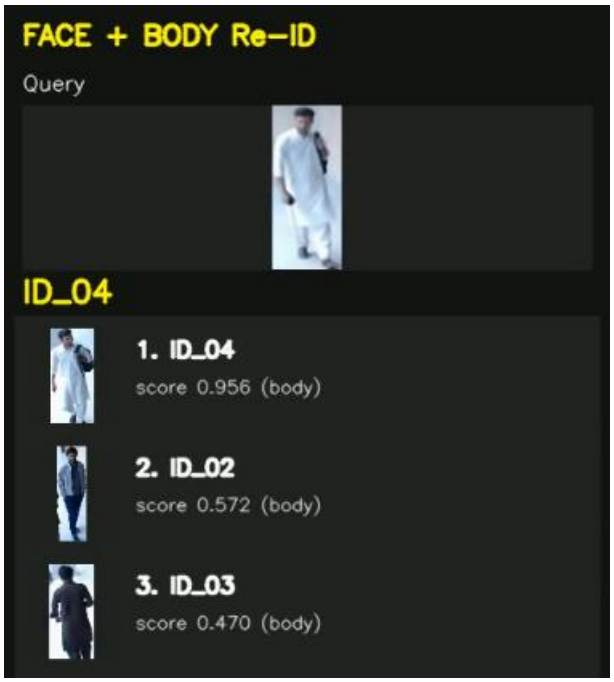


Figure 5.24: Rank-Based Matching Result Showing Top Identity Candidates and Similarity Scores.

The rank-representative display was helpful in understanding the identity decisions. In simple cases, the right ID always had a higher similarity score than the second choice. The Rank-1 and Rank-2 scores may become close in cases of low lighting or similar clothing. This revealed that the model was able to capture identities that are similar in appearance.

The result is significant as it gives explainability. It enables the user to find out if an identity match was accepted or rejected and why. The rank output was also very useful for debugging as errors of matching could be checked in the CSV log.

5.6 Face-Assisted and Supporting Appearance Results

Body appearance, based on OSNet, is the primary identity cue used in the system. Optional face-assisted information and supporting appearance features were employed as applicable. This design was employed as face information can be used for verifying identity but it does not always exist in crowd video.

In selected cases in which faces were visible, face-assisted support was found to be helpful. It was useful in cases where the body appearance was not completely reliable. But in most real-world situations, the face cue was not available, particularly in side-view, back-view, occlusion and low-light images. In these situations, the system continued to rely on body features and appearance cues based on OSNet.

Table 5.12: Face-Assisted Support Result Summary

Condition	Face Cue Available	System Behaviour
Clear front-view	Yes	Face cue used as optional supporting information
Side-view	No/Limited	System relied on OSNet body features only
Back-view	No	System relied on OSNet body features only
Low light	No/Unreliable	System relied on OSNet body features only
Occluded	No/Partial	System relied on OSNet body features only

5.7 Results Under Difficult Conditions

The system was tested under difficult conditions to verify the practicability of the robustness. The conditions were: occlusion, low light, similar clothing, side-view, back-view, and multiple people in one frame.

5.7.1 Similar Clothing

The differences between individuals were blurred by similar clothing. When this happens, the appearance of body features can become very similar and the highest ranked identity candidates may score very close to each other.

The system employed rank-based matching, margin checking, lower-body cues and optional face-assisted support to minimise wrong matching. However, similar clothing remained one of the most challenging circumstances.

5.7.2 Occlusion

Occlusion affected the quality of person crops. If a person was partially visible, then the extracted feature was less reliable since the whole body was not visible.

The system still processed despite mild occlusion. Heavy occlusion resulted in a higher possibility of track loss, poor feature extraction, and uncertain identity matching.

5.7.3 Low Light

Low light levels decreased image clarity and impacted detection and re-identification. In dim lighting, body features were less reliable, and face-assisted support was less available.

The system still worked on low-light videos, but the accuracy of the match was less than in the normal-light videos.

5.7.4 Face-Hidden and Back-View Cases

Cases were challenging due to the back view making the face difficult to see and the appearance of the body varying greatly from the front-view cases.



Figure 5.25: Back-View and Face-Hidden Result Showing Body-Based Matching Without Face Cue.

The outcome showed that the system was not inhibited when the face was not available. It persisted with the body appearance features and identity gallery based on OSNet.

Table 5.13: Major Result Trends Under Difficult Conditions

Difficult Condition	Impact on System	System Behaviour
Similar Clothing	Close similarity scores between different persons	Rank-based matching and margin checking applied
Occlusion	Partial crops, possible tracking loss	Crop quality check; mild occlusion handled
Low Light	Weaker features, less reliable crops	System continued; accuracy reduced
Back-View / Face-Hidden	Body appearance varies from front-view	OSNet body features used as primary cue

5.8 CSV Log Output Results

During the video processing, the system created CSV logs **(as shown in Figure 4.8).** These logs documented the identity of each frame and were used to confirm results beyond what could be observed from the annotated video alone.

The CSV log file provided support for result checking frame by frame since identity decisions could be verified. It was helpful in viewing the frame numbers, temporary track ID, assigned global ID, similarity score, ranking candidates, face availability, and bounding box information.

This output was necessary because the annotated video may not sufficiently convey the reason for the assignment of the identity. By using similarity scores and ranks of the candidates, the decision could be tracked in the CSV log.

5.9 Google Coral Edge TPU Feature Extraction Results

The Google Coral USB Accelerator was tested for the feature extraction part of the system using an Edge TPU compatible TensorFlow Lite model. The model path employed for testing was: models/person_reid_edgetpu.tflite

This test was designed to validate compatibility of the Coral TPU with feature extraction, beyond the CPU-only mode of inference.

The outcome verified the fact that the Coral TPU can be detected and adopted to the compatible feature extraction branch. Video input was provided and the system was able to detect human bodies, assign tracking IDs, extract human identity features using the Edge TPU compatible model and annotate the output video.

The result should be understood in the right context. YOLOv8 and PyTorch OSNet pipeline was not deployed directly on the Coral TPU. Detection and tracking remained in the core software and only the compatible feature extraction model in TensorFlow Lite was used with the Coral TPU. During hardware testing, some setup issues were noticed such as Python environment activation, NumPy compatibility, OpenCV compatibility, and selection of virtual environments. Once these problems had been addressed, the Edge TPU feature extraction branch was successfully run.

Table 5.14: Coral TPU Feature Extraction Result Summary

Metric	**Observation**
Coral USB Accelerator Detection	Successfully detected
Model Used	models/person_reid_edgetpu.tflite
Feature Extraction	Compatible TFLite branch successfully executed
Detection and Tracking	Remained in primary software (YOLOv8 + BoT-SORT)
Setup Issues	NumPy, OpenCV compatibility issues resolved
Full Pipeline on TPU	Not claimed; only feature extraction branch tested

5.10 Overall Assessment of the Implemented Solution

The overall results indicate that the implemented system has met the main project goals within the scope of the project. The system detected persons, tracked persons during continuous visibility, extracted OSNet-based features, matched returning persons with stored identities in the gallery, updated unique count, generated CSV logs, optionally supported face-assisted functionality, tested Top-10 query-gallery retrieval, and tested Coral TPU feature extraction.

Table 5.15: Overall Assessment Against Project Objectives

Objective	Assessment
Detect persons in crowd videos	Met: YOLOv8 successfully detected persons
Track persons during continuous visibility	Met: BoT-SORT provided short-term continuity
Extract appearance features using OSNet	Met: Fine-tuned model extracted reliable features
Match returning persons to gallery	Met in successful cases; challenging in difficult conditions
Minimise repeated counting	Met: Unique count stable on correct re-identification
Test Coral TPU edge extraction	Met: Compatible TFLite branch tested
Evaluate under various conditions	Met: 16 test cases across multiple conditions
Generate CSV result logs	Met: Frame-level logs generated for all test videos

5.11 Overall Result Trends

The results of the study revealed three main trends.

First, the quality of detection had an impact on all subsequent stages. Simply-cropped, well-lit frames were found to be more suitable for person cropping, tracking and re-identification. Crop quality was affected by poor lighting and occlusion, and frame-edge positions also affected matching.

Second, appearance quality was a very important factor in re-identification. Matching of identities was more stable when the person's body was clearly visible and when clothing was distinct. Similarity scores were not as reliable when the person came from a different direction, in low lighting, or when only a portion of their body was visible.

Third, face-assisted support was helpful only when a clear face was present. For front-view cases, face data might be used as a trait in identity matching. The system relied naturally on body features and appearance cues from OSNet in the cases of side-view, back-view, low-light, and occluded cases.

Table 5.16: Summary of Major Result Trends

Trend	Observation
Detection quality drives pipeline	Clear well-lit crops improve all subsequent stages
Appearance quality matters	Distinct clothing and clear body view = stable matching
Face cue is supplementary	Useful when visible; system works without it

5.12 Results Summary

We evaluated the implemented Human Re-Identification Using Machine Learning system using the major test cases as outlined in Chapter 4. The outcomes demonstrated that the system is able to carry out the principal functions needed for person re-identification and unique counting.

The system automatically detected people with YOLOv8, tracked those who could be seen with BoT-SORT, extracted appearance features based on OSNet, compared identities with cosine similarity and rank-based matching, assigned global identities with an identity gallery, and created unique person counts. The CSV logs were used as further evidence to verify identity decisions and similarity scores.

The model performance was facilitated by the training and the evaluation in the query-gallery. The training loss reduced and stabilized after 60 epochs, and CMC accuracy and mAP demonstrated good query-gallery matching on the prepared project dataset. The Top-10 query-gallery retrieval test also confirmed the ability of the trained OSNet-based model in feature matching.

The best results were obtained with normal illumination, clear visibility of the body and distinct clothing. Low light, occlusion, similar clothing, side-view and back-view were the most challenging cases. Face-assisted support was enabled in some cases where a face was visible, but the system was able to correctly continue using body-based OSNet features when the face cue was not available.

The Coral TPU test was successful, confirming that it could be used for compatible extraction of TensorFlow Lite features. The entire YOLOv8 and PyTorch OSNet pipeline was not claimed to run directly on the Coral TPU.

Chapter 6

Discussion

An objective discussion of the results of the implemented Human Re-Identification Using Machine Learning system is provided in this chapter. The methodology, testing and results presented in the preceding chapters are used to discuss the results. No new methods, test cases, or results are introduced in this chapter.

The five main points of discussion are: what the project aimed to do, what actually happened in the testing process, why the results are important, why it did not work out as desired in the best-case scenario, and what could not be accomplished within the project's scope.

6.1 Original Problem and Project Aim

Repeated counting was the original problem being investigated in this project with regard to crowd monitoring. A normal person detection system can detect a person in video frames but cannot remember whether the same person was found before or not. Likewise, a tracking system can track a person while that person is always visible but can give a different temporary ID to that person if he or she moves outside of the camera's line of sight and comes into view again.

This results in a repeated counting issue. For instance, if a person enters a monitored area, then goes out of the view of the camera and back in again later, a detection-only or tracking-only system will record the person more than once. This makes the count of the crowd less reliable.

The purpose of this project was to create an appearance detection and tracking system capable of continuous appearance detection, extraction of appearance features, re-identification of the persons, and generation of a unique person count rather than a repeated appearance count. The goal was not to realise ideal person re-identification in all possible real-world scenarios. Rather, the goal was to show that a re-identification-based method can minimise repeated counting under recorded multiple-angle video conditions where the camera is the same.

6.2 Discussion of Main Results

The outcomes indicated that the system implemented has accomplished the major functions required for the project. The system identified individuals with YOLOv8, temporarily tracked them with BoT-SORT, extracted appearance features through an OSNet-based model, matched identities with cosine similarity and rank matching, and stored the identities that had been previously seen in an identity gallery.

The detection was better when people were visible and lighting was normal. Clear detections produced better person crops, aiding the subsequent tracking and re-identification phases. This demonstrates that quality detection can have a significant impact on the entire pipeline. With weak detection, the quality of the crop is weak, and the feature extractor might not generate a good identity feature.

BoT-SORT gave useful short-term tracking during times when the people are still discernible between consecutive frames. The results also revealed why just keeping track was not sufficient. If a person left the frame or became hidden or returned after a time delay, the temporary tracker ID may change. This verified the need for the re-identification phase.

The identity memory in the system was primarily the re-identification module based on OSNet. If no other person in the gallery had a similar appearance and the returning person's appearance was sufficiently similar, the system could re-assign the global ID that had been assigned to them in the past. In these instances, the unique count did not increase when it was not needed. This is directly related to the main objective of the project.

The system was also more interpretable due to the rank-based matching. The system gave top-ranked identity candidates and similarity information instead of just the ID. This facilitated understanding of the reasons for matching or for the creation of a new identity. Decisions were more confident when the best match was clearly higher than the next candidate. For nearest scores, output revealed uncertainty regarding the identity decision.

The query-gallery evaluation also validated model training. The training loss reduced and stabilized after 60 epochs and the CMC accuracy and mAP performed well on the prepared project dataset. The query set consisted of 35 images; hence the results are project-specific performance under same-camera multi-angle conditions and not a universal benchmark result.

6.3 Discussion of Face-Assisted Support

As the primary identity cue, the system relies on the body appearance with OSNet. If a clear face was provided, face-assisted information was provided only as an optional supporting cue.

This design was suitable as faces are not always visible in crowd videos. The person can be viewed from the side or back, partially hidden, or taken in low light. Consider an extreme case where the system was only aware of facial information; it would not work in many real-world situations. So it is better to keep the body-based OSNet features as the primary identity cue, thus making the system more reliable.

The results indicated that face-assisted support could be beneficial in certain scenarios in which a face is clearly visible. It might be useful when clothing was similar and body appearance alone was not distinctive enough. In side-view, back-view, low-light or occluded situations, however, the face cue was limited or not available. In those cases, the system automatically continued to recognise based on visual information and body appearance.

This implies that the face-assisted component is not the project's primary contribution; rather it is a means of support.

6.4 Discussion of Coral TPU Feature Extraction

The Coral TPU test verified that the system can be powered by a model capable of feature extraction using TFLite. The result is significant because it shows that some of the re-identification pipeline can be pushed closer to local edge hardware.

It is important to correctly interpret the Coral TPU result, however. It is not the whole YOLOv8 and PyTorch OSNet pipeline that was deployed on the Coral TPU. The compatible TFLite feature extraction branch was the only branch that used the TPU. Detection and tracking continued to be part of the primary software stream.

This is still useful as feature extraction is an important component of a re-identification system. Edge hardware can be used to alleviate reliance on cloud processing and could enable local deployment in event or crowd-monitoring environments in future releases. The project is still a partial edge-deployment project.

6.5 Implications of the Project

The primary implication of the project is that re-identification-based crowd counting reduces repeated counting and improves crowd counting accuracy. The normal counting system keeps track of how many times an object has appeared; the proposed counting system tries to keep track of how many unique persons have appeared by tracking the identity of previously seen persons.

This comes in handy at various events, like at universities, exhibitions, seminars and public gatherings. There may be many entries and exits made by people in such environments. In situations where the same individual is counted more than once, an accurate picture of attendance is not obtained. A unique counting system could give better information for seating, security, resource planning and crowd management.

Another implication is the need for explainable outputs. The system output can be reviewed after the system has processed the logs and rank-based matching data stored in the CSV. This is helpful because person re-identification is seldom foolproof. Similarity scores, rank candidates and frame-level info can be used to determine why a wrong match has happened.

The project also demonstrates the need for more than one method to achieve a complete re-identification solution. Detection, tracking, feature extraction, gallery storage, similarity comparison, rank matching, and logging are all interconnected processes. Once one stage is vulnerable, the later stages will be impacted.

6.6 Limitations of the Project

The project was successfully completed within the scope, but some limitations were encountered.

The first limitation is that the system was evaluated on recorded videos with multiple angles on the same camera, not in full multi-camera live deployment. Hence, the project cannot be considered as a completely functioning multi-camera re-identification system.

The second constraint was that the dataset was not very large compared to large public re-identification datasets. A total of 14,200 usable cropped person images from over 30 individuals were used in the project. This was sufficient for project-level fine-tuning and evaluation but still small in comparison to large datasets.

The third drawback is that the query-gallery evaluation is too small for broad interpretation. The query set was comprised of 35 images. The CMC and mAP are high, indicating good performance on the prepared project set, however due to the small number of queries, the results may not be interpreted as broadly as on other larger sets.

The fourth restriction is the strong dependence of performance on the quality of the crops. If YOLOv8 gives an incomplete bounding box or the person is blurred, dark or partly hidden, the OSNet feature may be weak. Weak crops can decrease matching accuracy.

The fifth restriction is similar attire. Similar-looking outfits can generate close similarity scores when two different individuals wear them. This complicates the process of identity matching.

The sixth restriction is the visibility of the faces. In some instances, face-assisted support can help, but often there are no faces. Side-view, back-view, low-light or occluded frames may not contain a face cue.

The seventh limitation is Coral TPU compatibility. Only models compatible with the Coral TPU can be accelerated. It cannot directly execute the entire PyTorch OSNet and YOLOv8 pipeline. Hence, the hardware aspect of the project is limited to feature extraction compatibility only.

6.7 Failings and Their Reasons

The project was not successful in achieving the best-case scenario in some parts.

In all challenging cases the system could not match the images perfectly. The primary reason is that person re-identification relies on visual similarity. When the amount of visual information is low (such as when wearing occluded or back-view appearance), or when there is visual confusion, the information available in the visual system is weak. This results in a less reliable feature vector.

Heavy occlusion was not completely addressed in the system. If the person is strongly blocked by another person or object, the detector may produce an incomplete crop, or the tracker might lose the person. The system can continue in the presence of mild occlusion but heavy occlusion is a challenging condition.

Multi-camera deployment was not completed. The reason is that multi-camera re-identification demands multiple synchronized camera streams, identification testing for

each camera viewpoint, and identity comparison across various camera viewpoints. This was outside of the scope and testing configuration implemented.

The system was not able to deploy the full pipeline on Coral TPU. Hardware and model compatibility is the reason. The models have to be quantised and compiled using TensorFlow Lite, and the main YOLOv8 and PyTorch OSNet pipeline will not work directly on Coral TPU. Consequently, the TPU was only tested with the compatible feature extraction branch.

The system needed threshold tuning as well. The thresholds used in person re-identification have an impact on whether the system generates a new ID or a recognised one. Too low a threshold may combine different people, and too high a threshold may not be able to re-identify the same person. A complete automatic threshold selection method was not developed.

6.8 Discussion Summary

The results of the implemented Human Re-Identification Using Machine Learning system were discussed in this chapter with respect to the original problem. The goal of this project was to decrease repeated counting by detecting, tracking, extracting appearance features based on OSNet and matching returning individuals to a record of identities and updating a unique count.

The system was found to be well suited for normal lighting, clear body visibility and distinct clothing conditions. The system could successfully re-identify a person who was trying to be counted a second time in successful cases, and could minimise double counting by using the identity gallery. The identity decisions were made more transparent by the implementation of ranked matching and CSV logs.

The project also demonstrated the practicability of crowd monitoring, as a unique person count is more beneficial than a mere count of appearances. The system remains imperfect. Performance was lower with low light levels, heavy occlusion, similar clothing and back-view cases. The work assumed recorded videos on the same camera and was not a full multi-camera implementation. The full pipeline of YOLOv8 and PyTorch OSNet was not used with Coral TPU; only compatible feature extraction was tested.

In general, the discussion reveals that the main objective of the system was fulfilled within a controlled and honest project scope, and some limitations as well as areas for future improvement were identified.

Chapter 7

Conclusion

We solved this problem of repeated counting in crowd monitoring by developing a Human Re-Identification Using Machine Learning system. When using simple detection-based counting, an individual can be counted again when he or she exits the camera vision and returns later. Likewise with a tracking-only system, a new temporary ID will be issued after occlusion, disappearance, or re-entry. Thus, this project aimed to determine who was seen before and to keep a more accurate Unique Person Count.

The solution implemented integrated a person detector using YOLOv8, a short-term tracker using BoT-SORT, a feature extractor using OSNet, a model for comparing identities based on cosine similarity, a model for deciding the identity of a person based on their rank, and an identity gallery for storing people for whom the detector has already seen. A total of 14,200 usable cropped person images was taken from over 30 people, with 11,335 of these images being used for fine-tuning the OSNet-based model. The remaining dataset consisted of 2,830 testing/gallery images and 35 query images for query-gallery evaluation.

The appearance cues were only provided when possible, such as face-assisted and supporting appearance cues, with the primary decision made based on body appearance features. The system was tested with over a dozen videos recorded in a variety of conditions such as normal light, low light, occlusion, similar clothing, front, side, back views, person leaving and returning, and multiple people in one frame.

The results indicated that the system was able to detect people, assign global identities, re-identify people if they had returned in successful cases, update the unique count, and produce CSV logs for verification. The training loss, CMC accuracy, and mAP results also confirmed that the identity learning features were useful for the prepared dataset using the fine-tuned OSNet-based model.

Google Coral USB Accelerator was also tested for the compatible feature extraction part using TensorFlow Lite. This facilitated partial edge-based inference, but it was not stated that the entire YOLOv8 and PyTorch OSNet pipeline would run directly on the TPU.

In general, the project was able to realise a same-camera multi-angle human re-identification and unique counting system. It solved the problem of repeated counting by using the identity gallery matching in addition to detection or temporary tracking ID. While the system still has limitations in low light, occlusion, similar clothing and back-view scenarios, it still offers great promise for real-world crowd monitoring use cases, where counting individuals rather than repeated instances is more beneficial.

Chapter 8

Future Work

The system built in this project demonstrates human re-identification and unique person counting in the same-camera multi-angle recorded-video scenario. There are several modifications that can be implemented in the future to make this system more robust, scalable and feasible for real-world applications.

8.1 Multi-Camera Re-Identification

Videos were recorded using the same setup with various angles and tested in the present project. The system could be expanded to multi-camera re-identification in the future, allowing the same person to be identified across different cameras within a larger monitored area.

Multiple cameras would be needed at various places to develop this direction. There would also need to be camera synchronization, camera IDs, timestamps and an identity gallery that all cameras share. Another large multi-camera dataset is also required to make the model learn from appearance changes in different camera views.

8.2 Real-Time Live Video Processing

Recorded videos of the current system were used for the evaluation. The system can be improved in the future to process camera feed in real-time live mode. This would enable the unique count to be incremented continuously over the course of an event.

This would need more powerful local hardware, faster inference speed, efficient frame buffering, and a shorter processing delay. Better memory management for the identity gallery would be required for a real-time version, as the gallery would need to update continuously with the video stream.

8.3 Larger and More Diverse Dataset

The prepared project dataset was used to fine-tune the OSNet-based model. The dataset is helpful for this project but a larger and more diverse dataset would be beneficial for future research. More people, more variations of clothing, more lighting, more body orientation, more camera heights, and more crowded scenes would all improve the system.

This would involve increased data collection sessions, correct labelling of the identities, storage space and time for cleaning and organising the dataset. More data would yield better generalisation of the system to new environments and minimise overfitting to the existing testing environment.

8.4 Improved Robustness in Difficult Conditions

The results indicated that low light, occlusion, similar clothing, back-view appearance and hidden faces were the most challenging conditions. Future work is needed to optimise performance in such challenging cases.

It can be achieved via low-light image enhancement, enhanced crop quality verification, part-based body feature extraction, improved feature matching for lower body, and additional training samples from challenging conditions. Face-assisted support can be enhanced, but should not be mandatory, since faces are not always seen in crowd videos.

8.5 Adaptive Threshold and Matching Strategy

In the present system, the similarity thresholds and margin conditions were tuned during testing. The system can be extended in future by incorporating the ability to adapt the threshold value to match the light condition, crowd density, confidence score, and the quality of features.

This would need a calibration dataset and further experimental testing. Wrong identity merging could be minimised by adaptive matching when individuals look alike and false new IDs could be minimised when a user returns with a different angle or lighting condition.

8.6 Stronger Edge Deployment

The Google Coral TPU was used to test the compatible feature extraction part of the system using TensorFlow Lite. The edge deployment can be enhanced in the future by making more components edge compatible.

This would involve model quantization, TensorFlow Lite conversion, Edge TPU compilation and hardware testing. Another potential variation could be multiple edge devices or a more powerful embedded platform. This will alleviate reliance on cloud processing and make it more useful for in-person crowd monitoring.

8.7 Future Work Summary

In general, future research is needed to expand the project from same-camera multi-angle testing to full multi-camera re-identification, enhance the robustness in challenging visual scenarios, increase the number and diversity of the dataset, enhance the real-time video processing, select the threshold adaptively, and strengthen the deployment of edge hardware. This improvement would make the system more useful in practical crowd management applications in which the lighting, the density of the crowd, the position of the camera, and the movement of the persons vary continuously.

Chapter 9

Reflections on Learning

9.1 Overall Learning

One of the most comprehensive engineering learning experiences we had during our undergraduate careers was this project. It enabled us to tackle a real problem, with a clear practical goal. Instead of just finishing a theoretical problem, we had to learn how to solve the problem, gather data, create a model, test the system, discover fails and make the system better and better over time.

The main knowledge gained from this project was an understanding of a real world engineering problem and how this could be transformed into an effective technical solution. Initially human re-identification seemed to be a straightforward counting task. But in the implementation process we found that it's not sufficient to detect a person, it's also necessary to know whether that person has been detected before. This enabled us to understand the importance of integrating detection, tracking, feature extraction, similarity matching and identity gallery management into a single system.

9.2 Technical Skills Developed

The project was instrumental in learning a number of technical capabilities related to computer vision, machine learning, deep learning, video processing, and the deployment of AI on hardware. We have introduced the YOLOv8 model for person detection, the BoT-SORT model for short-term tracking, and an OSNet model for person re-identification. We also learned how to process videos frame by frame, crop the persons detected, extract the identity features, compare the feature vectors, assign the global IDs and generate the annotated videos and CSV logs.

By fine-tuning our own dataset to the OSNet-based model, we gained insight into the process of dataset preparation and model training. We understood the significance of clean person crops, organize data by identifying, splitting training and testing, and query-gallery evaluation. The debugging process also helped us to better understand the technical aspects as an error may be due to detection, tracking, feature extraction, face assisted support, similarity threshold or the logic of the gallery update.

The Google Coral USB Accelerator opened up a new way of using edge AI for us. We learned that it is not only about connecting hardware. It also needs to have compatible model formats, TensorFlow Lite conversion, compilation for Edge TPU, and the right Python environment and library versions. As electrical engineering students, this enabled us to relate artificial intelligence, software implementation and hardware-based processing.

9.3 Non-Technical Skills Developed

The project also trained valuable non-technical skills. Time management, consistency and patience and self-discipline were necessary to plan and complete a multi-stage system. The project has several steps, so we need to split it into several steps, test each step, and merge all the steps into one step of pipeline.

The skills of communication were also enhanced through the process of report writing. We had some training on communicating technical work clearly and systematically. We gained insight into the need for clear and honest technical writing by creating the methodology, testing, results, discussion, and conclusion chapters. We also learned the limitations must not be hidden; it demonstrates genuine grasp of the project.

Another key learning outcome was team work. Coordination and communication was needed to divide responsibilities, link together various parts of the code, check results, prepare figures and keep the report consistent. Also socialization on problem discussion and accepting criticism and improvement of the project based on the supervision.

9.4 Personality Development

The project helped us to gain confidence, patience and engineering maturity. The challenge of human re-identification in low-light, occluded, similar clothing, and different viewing angles is not an easy task. The system was not completely successful in all conditions from the start. We needed to test the algorithm multiple times, correct errors, change the thresholds, optimize the pipeline and have to comprehend the reasons for some cases being more difficult than others.

The project had taught us that real engineering systems don’t always work exactly as they’re supposed to. The good engineer should not only demonstrate good success but also recognize failures and provide an explanation as to why. This experience developed the

problem solving attitude and strengthened our confidence to solve the unknown technical problems.

The project also helped us to develop our logical thinking skills. We found that we didn't have to consider the system as a whole, but rather divide it up into smaller steps: detection, tracking, feature extraction, matching, counting, logging, and hardware testing. This was a thinking-by-the-same-sudden-as-the-light switch approach that has aided us in problem solving.

9.5 What Could Have Been Improved

In retrospect there is a number of things that could have been improved with more time and resources. With a larger and more diverse set of data, the generalization of the OSNet-based model could be further improved. The more identities, the more clothing styles, the more lighting conditions, and the more crowded scenes, the better the model would have been.

Additional improvements to testing would have been made by incorporating live video rather than solely recorded video. If a live test had been performed, there would have been more of the challenges that occur in the real world, including processing delay, camera stability, and a constant updating of the identity gallery. In the future, it may be possible to add full multi-camera testing, which would need multiple cameras, synchronization and an identity gallery, but would be beneficial.

It may be better to have a hardware deployment part. In this project, the Coral TPU is only used for the feature extraction part of the TensorFlow Lite library that is compatible with the TPU. A future version of the pipeline might be capable of optimizing additional components to be deployed at the edges. This would take longer time, model conversion, quantization and hardware testing.

Overall, this project was beneficial to our technical and personal development. It enlightened us to computer vision, machine learning, video processing, edge AI, teamwork, technical writing and project management. Most importantly, it showed us how to approach an engineering problem with integrity, gradually refine it, and give honest explanations of its successes and failures.

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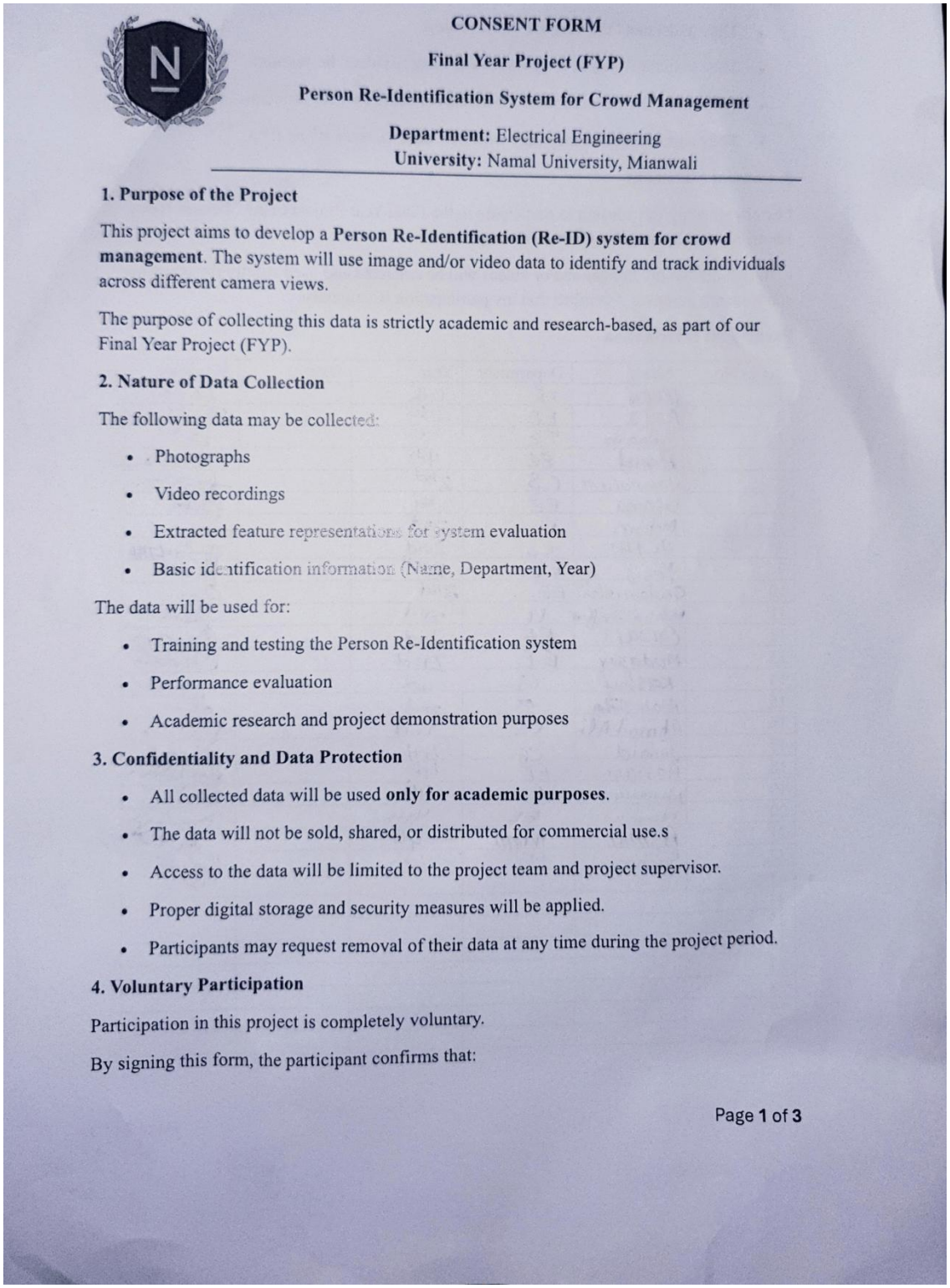
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Appendices

Appendix A

Concent Form



CONSENT FORM

Final Year Project (FYP)

Person Re-Identification System for Crowd Management

Department: Electrical Engineering
University: Namal University, Mianwali

1. Purpose of the Project

This project aims to develop a **Person Re-Identification (Re-ID) system for crowd management**. The system will use image and/or video data to identify and track individuals across different camera views.

The purpose of collecting this data is strictly academic and research-based, as part of our Final Year Project (FYP).

2. Nature of Data Collection

The following data may be collected:

- Photographs
- Video recordings
- Extracted feature representations for system evaluation
- Basic identification information (Name, Department, Year)

The data will be used for:

- Training and testing the Person Re-Identification system
- Performance evaluation
- Academic research and project demonstration purposes

3. Confidentiality and Data Protection

- All collected data will be used **only for academic purposes**.
- The data will not be sold, shared, or distributed for commercial use.s
- Access to the data will be limited to the project team and project supervisor.
- Proper digital storage and security measures will be applied.
- Participants may request removal of their data at any time during the project period.

4. Voluntary Participation

Participation in this project is completely voluntary.

By signing this form, the participant confirms that:

Page **1** of **3**

- They understand the purpose of the project.
- They voluntarily agree to provide their images/videos for research.
- They allow the research team to use the collected data for academic purposes.
- They understand that they may withdraw their consent at any time.

5. Consent Statement

I hereby give my full consent to participate in the Final Year Project titled **"Person Re-Identification System for Crowd Management."**

I understand that my images and/or videos will be collected and used strictly for academic and research purposes. I confirm that my participation is voluntary.

Participant Information

Serial No	Name	Department	Year	Date	Signature
1	IRFAN	EE	4th		
2	AZIZ	EE	4th		
3	Sohaib	EE	4th		
4	Asmat	EE	4th		
5	Hassnain Ali	CS	2nd		
6	Usama	EE	2nd		
7	Mohsin	EE	2nd		
8	ALTAF	CS	2nd		
9	Yasir	EE	2nd		
10	Ghulam Ullah	EE	2nd		
11	Mohsin Ghaffar	EE	2nd		
12	Owais	EE	2nd		
13	Mudasar	EE	2nd		
14	Kashif	CS	2nd		
15	Allah Ditta	CS	2nd		
16	Ahmad Ali	CS	2nd		
17	Junaid	CS	4th		
18	Hassam	EE	4th		
19	Munawar	CS	4th		
20	Noman	EE	4th		
21	Asmat	Math	4th		
22	Shahmeer	EE	4th		
23					
24					
25					
26					
27					
28					
29					
30					

FYP Team Information

Project Students:

- Muhammad Wajid Jamal
- Muhammad Usman Shah

Supervisor:
Dr. Mudassar Raza
Co- Supervisor:
Dr. Farrukh Qureshi

Appendix B

Use of AI Help

During this project, we used AI tools such as ChatGPT and Claude for help in learning and report writing. We did not use AI to do the full project for us. The project idea, coding, testing, videos, screenshots, results, and final decisions were done by our team.

AI was used mainly to improve our writing, grammar, sentence flow, and understanding of some technical concepts. We also used it to check whether our report sections were clear, connected, and suitable for a final year project report.

B.1 Where AI Was Used

AI was used for the following support:

- to understand some concepts in computer vision and machine learning
- to improve grammar and writing flow.
- to make paragraphs more clear and professional.
- to check if the report section was connected with our project.
- to improve headings and structure.
- to check if any sentence was overclaimed.
- to make the writing simple and easy to understand.

AI was not used to create fake results, fake screenshots, fake accuracy, or fake testing. All results and screenshots in this report are from our own project work.

B.2 Example Prompts Used

The following are some example prompts that show how AI was used.

Prompt 1: Introduction improvement

This is my introduction section. Please improve the grammar, sentence flow, and writing style.

Keep the meaning same and make it suitable for a final year engineering report.

Prompt 2: Literature review improvement

Please improve this literature review section. Make it more connected with my project and remove repeated lines. Keep the language simple and professional.

Prompt 3: Understanding human re-identification

Explain human re-identification in simple English. Also explain how it is different from person detection and tracking.

Prompt 4: Understanding models

Explain how YOLOv8n is used for person detection and how ByteTrack helps in tracking people in video.

Prompt 5: Understanding feature matching

Explain cosine similarity in simple words and how it is used to compare two person feature vectors.

Prompt 6: Methodology improvement

This is my methodology section. Please improve the wording and flow according to the required headings. Do not add anything that is not part of my project.

Prompt 7: Testing chapter

Write this testing section in a clear way. Only explain what was tested, why it was tested, and how it was tested. Do not discuss results in detail.

Prompt 8: Results chapter

Improve this results section. Focus on major trends and supporting figures. Do not invent exact numbers or unsupported results. **Prompt 9: Discussion chapter**

Improve this discussion section. Discuss the original problem, results, implications, limitations, and failings. Do not add new methods or new results.

Prompt 10: Conclusion improvement

This is my conclusion. Make it short, clear, and connected with the introduction. Do not add new information.

Prompt 11: Grammar checking

Check this paragraph grammatically and improve the wording. Keep the meaning same and use simple English.

Prompt 12: Overclaim checking

Check this paragraph and tell me if any sentence is overclaiming. Make it honest according to my actual project.

B.3 How AI Help Was Checked

We did not copy AI answers directly without checking. We reviewed the answers and changed them according to our actual project. For example, we checked model names, tracker name, testing conditions, video details, and hardware details before adding them to the report.

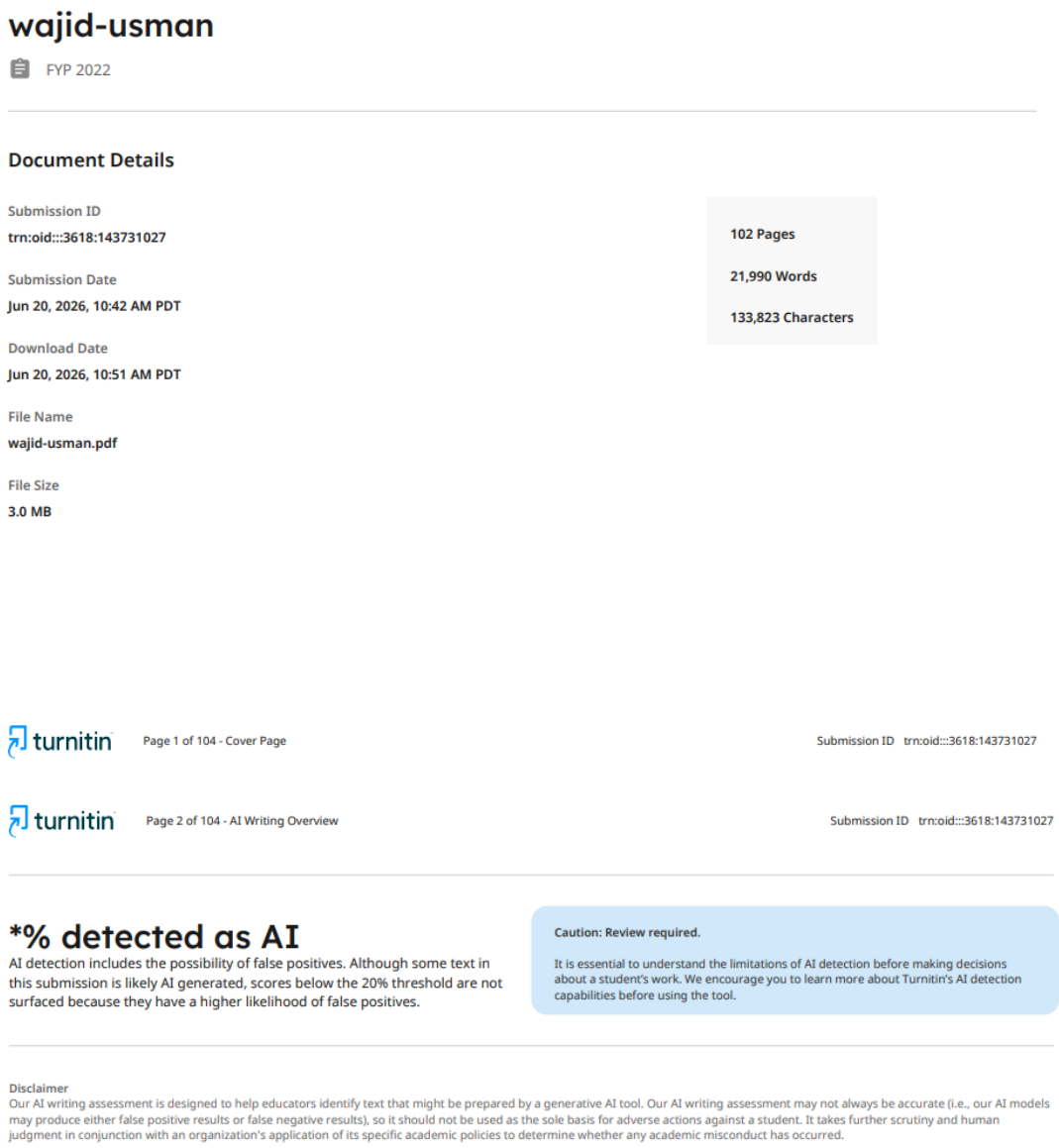
If any AI suggestion did not match our project, we removed it or corrected it. The final report is based on our own implementation, testing, screenshots, results, and understanding.

B.4 Final Statement

AI was used only as a support tool for learning, grammar correction, writing improvement, and report organization. The actual engineering work, including coding, testing, video processing, result collection, screenshots, and final explanation, was completed by our team. The use of AI is mentioned here for honesty and transparency.

Appendix C

Turnitin Report (Similarity + AI)



wajid-usman

FYP 2022

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4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

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Matches with neither in-text citation nor quotation marks

8 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

2% Internet sources
1% Publications
4% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.